Branch Selection and Recursive Complex Root Dynamics

Principal Orbits, Symbolic Itineraries, Inverse Trees, Monodromy, Geometric Limits, and Branch Measures

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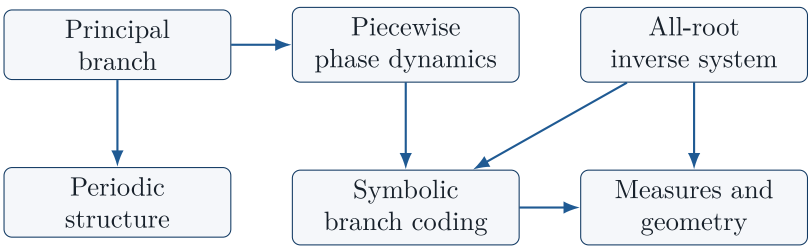
Self-contained analytical preprint

A systematic study of branch-sensitive recursion generated by $z_{n+1}^r = -z_n$, with the principal and multivalued viewpoints kept distinct.



Abstract

Repeated extraction of complex roots is naturally multivalued. A deterministic iteration arises only after a branch-selection rule is specified, whereas the complete root operation is more naturally represented as inverse dynamics. This paper studies the family

$$w^r = -z, \qquad r \geq 2,$$

from both viewpoints and develops a unified description while preserving the mathematical distinctions between them.

For the principal branch

$$F_r(z) = \mathrm{Root}_r^{\mathrm{pr}}(-z),$$

with principal argument in $(-\pi, \pi]$, the radial and angular dynamics are exactly solvable. Every nonzero orbit approaches a unique attracting 2-cycle, and there are no nonzero periodic orbits of any other period. The mechanism is unusually transparent after separating logarithmic radius from a signed principal angle: the radial variable contracts by $1/r$, while the absolute angular variable obeys the affine contraction $\alpha \mapsto (\pi - \alpha)/r$ with reversed orientation.

A phase-perturbed family
$$F_{r,\beta}(z) = \mathrm{Root}_r^{\mathrm{pr}}(e^{i\beta}z)$$
is analyzed as a piecewise-affine principal-angle system. Exact symbolic formulas are obtained for periodic itineraries, and endpoint-sensitive admissibility conditions are written explicitly. A natural rescaling of the post-root invariant angular interval gives an exact conjugacy with the standard contracted rotation $x \mapsto \{x/r + \delta\}$, where $\delta = (r-1)/(2r) - \beta/(2\pi) \pmod 1$. This makes the asymptotic complex phase set an explicit image of the contracted-rotation attractor. In particular, the paper derives explicit period-one and period-two mode-locking windows, including the fact that the original $\beta = \pi$ negative-root case sits at the center of a robust period-two interval.

The full multivalued root operation is the inverse correspondence of the monomial $P_r(z) = -z^r$. Its depth-n inverse level is an explicitly rotated regular r^n-gon. These polygonal levels converge in Hausdorff distance to the unit circle, and an exact Hausdorff formula yields a sharp first-order asymptotic constant. The associated finite and infinite branch structures admit base-r symbolic coding; through linear conjugacy with z^r, the tree is connected with the classical rooted-tree and iterated-monodromy picture.

Finally, probability measures induced by branch selection are treated at the symbolic level. Uniform selection gives Haar measure on the circle, while iid biased selection gives Bernoulli self-similar measures with the exact entropy–dimension relation
$$\dim_H \nu_p = \frac{-\sum_{k=0}^{r-1} p_k \log p_k}{\log r}.$$
The paper also records a variable-order extension, an explicit logarithmic branch-defect formalism for principal powers, and a set of carefully separated open problems. The central mathematical contribution is the explicit, fully computed synthesis of these structures in the negative-root monomial model, especially the exact contracted-rotation realization of the phase family and the resulting global description of its complex asymptotic sets.

Keywords. Complex roots; multivalued functions; principal branch; complex dynamics; contracted rotations; rotation numbers; arithmetic dynamics; piecewise-affine phase dynamics; inverse iteration; symbolic dynamics; iterated monodromy; regular polygons; Julia sets; potential theory; Hausdorff convergence; self-similar measures; entropy; dimension; random dynamics.

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Core message: the same recursive radical gives different limiting objects depending on whether one fixes a branch, retains all branches, or assigns a probability law to the branches.

Contents

1 Introduction

A complex root is not, in general, a single-valued operation. For $z \neq 0$ and an integer $r \geq 2$, the equation

$$w^r = z$$

has precisely r distinct solutions. Consequently, an expression such as "iterate the rth root" is mathematically incomplete until one has specified both a branch convention and a domain on which that convention is used.[1] More precisely, one specifies whether one means:

(i) the iteration of a chosen single-valued branch;

(ii) the iteration of a multivalued correspondence;

(iii) a random or otherwise non-deterministic branch-selection process;

(iv) inverse iteration of an associated polynomial map.

These objects are related, but they are not the same.

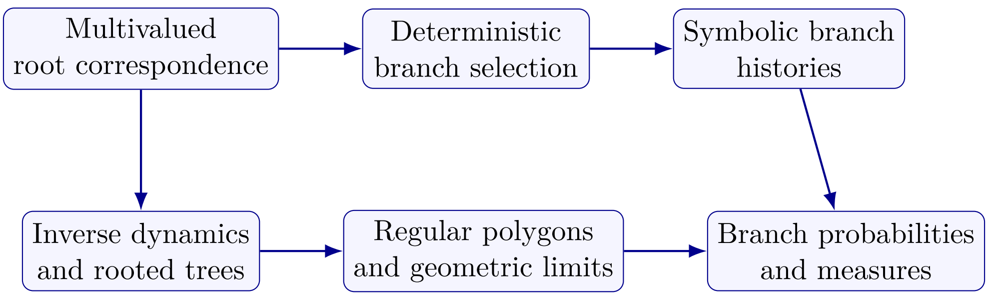
The motivating recurrence considered here is

$$z_{n+1}^r = -z_n. \tag{1}$$

At first sight, this is merely a recursive complex radical. However, once the distinction between branch selection and the full root correspondence is maintained, a surprisingly rich family of structures appears. Deterministic principal iteration leads to a completely solvable contracting dynamics. Allowing a phase parameter produces a piecewise-affine circle system. Allowing all roots produces an explicit inverse tree. Assigning probabilities to branches produces self-similar measures. The tree itself is naturally encoded by base-r words and is related to classical monodromy actions.

The purpose of this paper is not to claim that these surrounding subjects are new. Complex iterated radicals have a substantial history; see, for example, the work discussed by Walker [14], including earlier work of Ogilvy, Rohde, and Gerber. Contracted rotations and their rotation numbers are also well-established objects in dynamical systems [5]. The inverse dynamics of monomials and their rooted trees belong to classical complex dynamics and iterated monodromy theory [9, 10]. Likewise, the measure and dimension theory used later belongs to the standard theory of self-similar and symbolic measures [4, 2].

The contribution developed here is instead an explicit mathematical synthesis:



The central principle is simple but important:

[1] The point is not merely terminological: different branch selections can produce different dynamical systems from the same algebraic relation.

> A multivalued complex operation, a deterministic rule for choosing one of its branches, and the topology or dynamics of the complete family of branches should be studied as distinct mathematical objects.

The negative-root family is particularly suitable because all three levels can be analyzed explicitly.

1.1 A note on the development of the present work

The present paper develops and reorganizes a sequence of earlier investigations by the author concerning complex logarithms, complex powers, recursive negative roots, and inverse root structures. The earlier work on principal negative-root iteration motivates Section 4. The later inverse-tree formulation motivates Section 7–Section 11. Several formulations have been strengthened during the present development.

In particular, no statement in this paper should be read as identifying a principal branch with the full multivalued root correspondence. Furthermore, all statements involving the principal argument use the single convention fixed in Section 2. This convention is not cosmetic: half-open endpoint choices affect certain phase parameter intervals.

1.2 Structure of the paper

The paper proceeds from deterministic to multivalued structure.

1. Section 2 fixes the logarithmic and root conventions.
2. Section 4 solves the principal negative-root dynamics exactly.
3. Section 5 studies phase perturbations and their symbolic dynamics.
4. Section 6 identifies the phase dynamics exactly with contracted rotations and imports their arithmetic structure.
5. Section 7 develops the full inverse tree.
6. Section 8 relates branch words to rooted trees and monodromy.
7. Section 9 proves the geometric convergence to the unit circle.
8. Section 10 places the polygonal convergence in the framework of Julia sets, Green functions, capacity, and equilibrium measures.
9. Section 11 studies uniform and weighted branch measures.
10. Section 13 separates symbolic boundary measures from chronological random-root dynamics.
11. Section 14 gives a variable-order extension and its quantitative geometry.
12. Section 15 explains the logarithmic branch mechanism underlying complex-power identities.
13. Section 18 assembles the preceding results into one convergence-and-branch hierarchy.

2 Branch conventions and basic definitions

Throughout the paper, unless explicitly stated otherwise, the principal argument is

$$\operatorname{Arg} : \mathbb{C}^\times \longrightarrow (-\pi, \pi].$$

Thus every $z \in \mathbb{C}^\times$ has the unique representation

$$z = |z|\, e^{i \operatorname{Arg} z}, \qquad \operatorname{Arg} z \in (-\pi, \pi].$$

This half-open choice is fixed once and for all.[2]

Definition 2.1 (Principal logarithm) For $z \in \mathbb{C}^\times$, define

$$\operatorname{Log} z = \log |z| + i \operatorname{Arg} z.$$

The multivalued logarithm is

$$\log z = \{\log |z| + i(\operatorname{Arg} z + 2\pi k) : k \in \mathbb{Z}\}.$$

Definition 2.2 (Principal root) For $r \geq 2$ and $z \in \mathbb{C}^\times$, define

$$\operatorname{Root}_r^{\mathrm{pr}}(z) = \exp\left(\frac{1}{r} \operatorname{Log} z\right).$$

Set $\operatorname{Root}_r^{\mathrm{pr}}(0) = 0$.

The principal root is a single-valued section of the r-fold power covering only after the logarithmic sheet has been fixed.[3]

The full rth-root correspondence is

$$\mathcal{R}_r(z) = \{w \in \mathbb{C} : w^r = z\}.$$

It is useful to record immediately that

$$\operatorname{Root}_r^{\mathrm{pr}}(z) \in \mathcal{R}_r(z),$$

but

$$\operatorname{Root}_r^{\mathrm{pr}} \neq \mathcal{R}_r.$$

The first is a single-valued branch function, while the second is a set-valued correspondence.

Definition 2.3 (Negative-root correspondence and principal map) Fix $r \geq 2$. Define

$$\mathcal{F}_r(z) = \{w \in \mathbb{C} : w^r = -z\},$$

and define its principal branch

$$F_r(z) = \operatorname{Root}_r^{\mathrm{pr}}(-z).$$

The deterministic recurrence studied in Section 4 is

$$z_{n+1} = F_r(z_n).$$

The multivalued system studied later is

$$z_{n+1} \in \mathcal{F}_r(z_n).$$

Remark 2.4 The use of the same informal phrase "repeated roots" for both systems is potentially misleading. In the deterministic system every initial point has one orbit. In the multivalued system every nonzero point has r possible successors, producing an r-ary tree of histories.

[2] Changing the half-open endpoint changes formulas exactly at the branch cut; none of the intrinsic geometric sets change, but endpoint-sensitive symbolic statements can.

[3] This is why the operation cannot be treated as ordinary division of an angle on the quotient circle without retaining the branch-selection information.

3 Mathematical framework and a master structural theorem

Before beginning the detailed proofs, it is useful to state the hierarchy of objects that will be kept separate throughout the paper. The single equation

$$w^r = -z$$

has a finite set of solutions for each nonzero z, but choosing one of those solutions converts a correspondence into a function. The resulting function can then be iterated, while the original correspondence can instead be iterated as an inverse relation. These procedures produce different objects.

Structural guide

Four levels of the model. For fixed $r \geq 2$, the paper considers:

(i) the principal function $F_r(z) = \mathrm{Root}_r^{\mathrm{pr}}(-z)$;

(ii) the complete correspondence $\mathcal{F}_r(z) = \{w : w^r = -z\}$;

(iii) symbolic histories obtained by recording branch choices;

(iv) probability measures obtained by placing laws on those histories.

The first object produces a single orbit; the second produces a tree; the third records its combinatorics; the fourth records its statistical geometry.

The later sections are therefore not competing descriptions of one limit. They are descriptions of different mathematical constructions associated with the same algebraic relation. This distinction is the organizing principle of the paper.

Proposition 3.1 (Basic invariances). *For every $r \geq 2$, the principal map F_r has the following properties:*

(a) $F_r(0) = 0$;

(b) $|F_r(z)| = |z|^{1/r}$;

(c) $F_r(\lambda z) = \lambda^{1/r} F_r(z)$ *for* $\lambda > 0$;

(d) *on each region where the principal argument of $-z$ is represented by a fixed affine lift, the angular part is affine with slope $1/r$.*

Proof. The first statement follows from the definition at the origin. The second is proved in Section 4. The third is immediate from the logarithmic formula and positive real scaling. The fourth follows from the fact that $\mathrm{Arg}(-z)$ differs from a chosen angular lift by an integer multiple of 2π, after which division by r is affine. □

Theorem 3.2 (Master structural theorem). *Fix $r \geq 2$ and $z_0 \in \mathbb{C}^\times$. Then the recursion*

$$z_{n+1}^r = -z_n$$

has the following simultaneously valid descriptions.

(a) *Under the principal branch, $|z_n| = |z_0|^{r^{-n}}$, and the orbit has ω-limit set $\{e^{i\pi/(r+1)}, e^{-i\pi/(r+1)}\}$.*

(b) *Under full inverse iteration, the depth-n level consists of exactly r^n points forming a regular polygon of radius $|z_0|^{r^{-n}}$.*

(c) *These polygonal levels converge in Hausdorff distance to* $\mathbb{S}^1$.

(d) *Uniform weighting of the level converges weakly to Haar probability on* $\mathbb{S}^1$.

The constructions in (a)–(d) are compatible through the common radial law but are not identical limiting procedures.

Proof. Parts (a)–(d) are proved separately in Section 4, Section 7, Section 9, and Section 11. The point of stating them together is structural: the exact radial exponent r^{-n} is shared by the deterministic and inverse descriptions, while the angular operation changes from contraction of one selected angle to increasingly fine sampling of all angles. □

3.1 Notation at a glance

The following symbols recur throughout the paper.

Symbol	Meaning	Role
$\operatorname{Arg} z$	principal argument in $(-\pi, \pi]$	fixes the logarithmic sheet
$\operatorname{Log} z$	$\log\lvert z\rvert + i \operatorname{Arg} z$	defines the principal root
F_r	$\operatorname{Root}_r^{\mathrm{pr}}(-z)$	deterministic principal dynamics
$\mathcal{F}_r$	$\{w : w^r = -z\}$	full multivalued correspondence
P_r	$-z^r$	forward map whose inverse relation is $\mathcal{F}_r$
α_n	$\lvert\operatorname{Arg} z_n\rvert$	scalar angular state
$A_n(z_0)$	$P_r^{-n}(\{z_0\})$	depth-n inverse level
ρ_n	$\lvert z_0\rvert^{1/r^n}$	common fixed-order inverse-level radius
N_n	$\prod_{j=1}^n r_j$	cumulative degree in the variable-order model
$\mathcal{T}_r$	rooted r-ary history tree	symbolic branch structure
η_p	iid product measure on symbols	branch-source law
ν_p	pushforward of η_p to $\mathbb{S}^1$	limiting branch measure

Principal result

The exact core in one display. For the principal negative-root map,

$$|z_n| = |z_0|^{r^{-n}}, \qquad \alpha_{n+1} = \frac{\pi - \alpha_n}{r},$$

so

$$\alpha_n = \frac{\pi}{r+1} + \left(-\frac{1}{r}\right)^n \left(\alpha_0 - \frac{\pi}{r+1}\right).$$

For the full inverse correspondence,

$$A_n(z_0) = \left\{ |z_0|^{1/r^n} \exp\left(i\frac{\phi_n + 2\pi j}{r^n}\right) : 0 \le j < r^n \right\},$$

with $\phi_n = \operatorname{Arg}((-1)^{S_n} z_0)$ and $S_n = (r^n - 1)/(r-1)$.

4 Principal negative-root dynamics

We begin with the deterministic principal branch

$$F_r(z) = \mathrm{Root}_r^{\mathrm{pr}}(-z).$$

4.1 Continuity domains and the branch-cut jump

The principal logarithm is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$. Therefore $z \mapsto \mathrm{Root}_r^{\mathrm{pr}}(-z)$ is continuous and holomorphic away from the rotated cut $[0, \infty)$. The positive real axis is consequently not an incidental coordinate singularity: it is the geometric location where the selected branch jumps.

For $x > 0$, approaching the point x from below and above gives

$$\lim_{\theta \uparrow 0} F_r(xe^{i\theta}) = x^{1/r} e^{i\pi/r}, \qquad \lim_{\theta \downarrow 0} F_r(xe^{i\theta}) = x^{1/r} e^{-i\pi/r}.$$

Thus the two one-sided images differ by a rotation through $2\pi/r$. In the square-root case this is a half-plane jump through π, while for large r the two limiting directions become closer.

Proposition 4.1 (Sectorial local regularity). *On every closed angular sector avoiding the positive real ray, the map F_r is locally Lipschitz on $\mathbb{C}^\times$, and its angular component has derivative $1/r$ in the principal angular coordinate. There is no global Lipschitz extension to all of $\mathbb{C}^\times$: the branch jump obstructs global continuity across the positive real ray, while the derivative also becomes unbounded as $z \to 0$ because F_r has size $|z|^{1/r}$.*

Proof. Away from the cut, the principal logarithm is holomorphic, so the composite $F_r(z) = \exp((1/r)\,\mathrm{Log}(-z))$ is holomorphic. Holomorphicity gives local Lipschitz continuity on compact subsets. On each branch interval the angular formula is affine with slope $1/r$. □

Remark 4.2 The recurrence therefore contains two different kinds of behavior at once: contraction within each branch sector and a discrete jump when the orbit crosses the cut. The scalar α-recurrence isolates the first mechanism while retaining enough information to reconstruct the second through the alternating sign.

4.2 Radial dynamics

Proposition 4.3 (Exact radial recurrence). *For every $z \in \mathbb{C}$,*

$$|F_r(z)| = |z|^{1/r}.$$

Consequently, if

$$z_{n+1} = F_r(z_n),$$

then

$$|z_n| = |z_0|^{r^{-n}}. \tag{2}$$

Proof. If $z = 0$, the assertion is immediate. For $z \neq 0$,

$$F_r(z) = \exp\left(\frac{1}{r}\,\mathrm{Log}(-z)\right),$$

and therefore

$$|F_r(z)| = \exp\left(\Re\left(\frac{1}{r}\,\mathrm{Log}(-z)\right)\right) = \exp\left(\frac{1}{r}\log|z|\right) = |z|^{1/r}.$$

Iteration gives

$$|z_n| = |z_{n-1}|^{1/r} = \cdots = |z_0|^{1/r^n}.$$

□

Corollary 4.4 (Radial convergence). *If $z_0 \neq 0$, then*

$$|z_n| \longrightarrow 1.$$

Moreover,

$$\log |z_n| = \frac{\log |z_0|}{r^n}. \tag{3}$$

Thus every nonzero orbit is driven exponentially fast toward the unit circle in logarithmic radial coordinates.

4.3 The actual principal angular map

Let

$$z = e^{i\theta}, \qquad \theta \in (-\pi, \pi].$$

The principal argument of $-z$ is not simply $\theta + \pi$ without qualification; principal-value reduction is required.

Lemma 4.5 (Principal argument after multiplication by -1). *For $\theta \in (-\pi, \pi]$,*

$$\operatorname{Arg}(-e^{i\theta}) = \begin{cases} \theta + \pi, & -\pi < \theta \leq 0, \\ \theta - \pi, & 0 < \theta \leq \pi. \end{cases}$$

Proof. If $-\pi < \theta \leq 0$, then $\theta + \pi \in (0, \pi]$, already in the principal interval. If $0 < \theta \leq \pi$, then $\theta + \pi \in (\pi, 2\pi]$, whose principal representative is $\theta - \pi \in (-\pi, 0]$. □

Hence the principal angular map is

$$T_r(\theta) = \begin{cases} \dfrac{\theta + \pi}{r}, & -\pi < \theta \leq 0, \\ \dfrac{\theta - \pi}{r}, & 0 < \theta \leq \pi. \end{cases} \tag{4}$$

Remark 4.6 (Branch reduction is part of the dynamics) A formal affine map

$$\widetilde{T}(\theta) = \frac{\theta + \pi}{r}$$

on an unrestricted real coordinate is a different dynamical system from (4). The former is a lifted recurrence. The latter is the actual principal-value dynamics. The distinction is essential: principal reduction occurs after every multiplication by -1 and before every root is taken.

4.4 Signed-angle reduction

Define

$$\alpha = |\theta| \in [0, \pi].$$

For $\theta \neq 0$, (4) changes the sign of the angle and gives

$$|T_r(\theta)| = \frac{\pi - |\theta|}{r}.$$

The same formula holds for $\theta = 0$. Hence:

Proposition 4.7 (Scalar angular recurrence). *Let*

$$\alpha_n = |\operatorname{Arg} z_n| \in [0, \pi].$$

Then

$$\alpha_{n+1} = \frac{\pi - \alpha_n}{r}. \tag{5}$$

Proof. If $\theta_n \le 0$, then

$$T_r(\theta_n) = \frac{\theta_n + \pi}{r} \ge 0,$$

so

$$|T_r(\theta_n)| = \frac{\pi - |\theta_n|}{r}.$$

If $\theta_n > 0$, then

$$T_r(\theta_n) = \frac{\theta_n - \pi}{r} \le 0,$$

and again

$$|T_r(\theta_n)| = \frac{\pi - \theta_n}{r}.$$

□

The map

$$A_r(\alpha) = \frac{\pi - \alpha}{r}$$

is affine with slope $-1/r$.

Theorem 4.8 (Exact angular solution). *The recurrence* (5) *has the exact solution*

$$\alpha_n = \frac{\pi}{r+1} + \left(-\frac{1}{r}\right)^n \left(\alpha_0 - \frac{\pi}{r+1}\right). \tag{6}$$

In particular,

$$\alpha_n \longrightarrow \frac{\pi}{r+1}.$$

Proof. The unique fixed point α_* of A_r satisfies

$$\alpha_* = \frac{\pi - \alpha_*}{r},$$

hence

$$(r+1)\alpha_* = \pi, \qquad \alpha_* = \frac{\pi}{r+1}.$$

Subtracting α_* from the recurrence gives

$$\alpha_{n+1} - \alpha_* = -\frac{1}{r}(\alpha_n - \alpha_*).$$

Iteration yields (6). □

4.5 The attracting two-cycle

Define

$$\zeta_+ = e^{i\pi/(r+1)}, \qquad \zeta_- = e^{-i\pi/(r+1)}. \tag{7}$$

Theorem 4.9 (Unique nonzero attracting two-cycle). *For every integer $r \geq 2$,*

$$F_r(\zeta_+) = \zeta_-, \qquad F_r(\zeta_-) = \zeta_+.$$

Moreover, for every $z_0 \in \mathbb{C}^\times$,

$$\omega(z_0) = \{\zeta_+, \zeta_-\},$$

where $\omega(z_0)$ denotes the ω-limit set of the principal orbit.

Proof. The radial recurrence gives

$$|z_n| \to 1.$$

The scalar angular recurrence gives

$$|\operatorname{Arg} z_n| \to \frac{\pi}{r+1}.$$

Except at the instant when the angle is zero, the principal angular map changes the sign of the angle. Since

$$\frac{\pi}{r+1} > 0,$$

the asymptotic sign alternates. Therefore the even and odd subsequences converge to the two points in (7), in one order or the other.

Direct substitution verifies the cycle:

$$\operatorname{Arg}(-\zeta_+) = \frac{\pi}{r+1} - \pi = -\frac{r\pi}{r+1},$$

and division by r gives

$$-\frac{\pi}{r+1}.$$

Thus $F_r(\zeta_+) = \zeta_-$. The reverse relation is analogous. □

Corollary 4.10 (Exact convergence rate in signed magnitude). *For every nonzero orbit,*

$$\left| \alpha_n - \frac{\pi}{r+1} \right| = r^{-n} \left| \alpha_0 - \frac{\pi}{r+1} \right|.$$

4.6 Quantitative convergence in the complex plane

The angular formula and the radial formula can be combined into an explicit metric estimate. Let

$$\alpha_* := \frac{\pi}{r+1}, \qquad L := \log |z_0|.$$

After at most one step the sign of the principal angle alternates. Define $\varepsilon_n \in \{-1, +1\}$ for $n \geq 1$ by

$$\operatorname{Arg} z_n = \varepsilon_n \alpha_n.$$

Then $\varepsilon_{n+1} = -\varepsilon_n$, and the appropriate point on the attracting cycle is

$$\zeta_n := e^{i\varepsilon_n \alpha_*}.$$

Proposition 4.11 (Exact asymptotic distance to the attracting cycle). *For $z_0 \neq 0$,*

$$|z_n - \zeta_n|^2 = \left(|z_0|^{1/r^n} - 1\right)^2 + 2|z_0|^{1/r^n}\left(1 - \cos(\alpha_n - \alpha_*)\right) \tag{8}$$

for all $n \geq 1$. Consequently,

$$r^n |z_n - \zeta_n| \longrightarrow \sqrt{(\log|z_0|)^2 + \left(\alpha_0 - \frac{\pi}{r+1}\right)^2}. \tag{9}$$

Proof. The first identity is the cosine law applied to two complex numbers of radii $|z_0|^{1/r^n}$ and 1, whose angular difference is $\varepsilon_n(\alpha_n - \alpha_*)$. For the asymptotic statement, use

$$|z_0|^{1/r^n} = 1 + \frac{L}{r^n} + O(r^{-2n})$$

and

$$\alpha_n - \alpha_* = \left(-\frac{1}{r}\right)^n (\alpha_0 - \alpha_*).$$

Since $1 - \cos t = t^2/2 + O(t^4)$, equation (8) becomes

$$|z_n - \zeta_n|^2 = \frac{L^2 + (\alpha_0 - \alpha_*)^2}{r^{2n}} + O(r^{-3n}),$$

which gives (9) after taking square roots. □

Remark 4.12 (What converges, and what does not) The sequence (z_n) itself need not converge because the limiting point alternates between ζ_+ and ζ_-. The correct deterministic statement is that the distance to the *moving phase point* ζ_n decays like r^{-n}. Equivalently, the even and odd subsequences converge separately. This is stronger and more precise than merely stating that the ω-limit set is the two-cycle.

4.7 Worked examples: the square-root and cube-root cases

The smallest values of r illustrate the general formula without adding any new assumptions.

Example 4.13 (Square roots: $r = 2$) The recurrence is

$$z_{n+1}^2 = -z_n.$$

Its nonzero attracting cycle is

$$\left\{e^{i\pi/3}, e^{-i\pi/3}\right\}.$$

The angular magnitude satisfies

$$\alpha_{n+1} = \frac{\pi - \alpha_n}{2},$$

with solution

$$\alpha_n = \frac{\pi}{3} + \left(-\frac{1}{2}\right)^n \left(\alpha_0 - \frac{\pi}{3}\right).$$

The inverse level $A_n(z_0)$ has 2^n vertices, and its angular mesh is $2\pi/2^n$.

Example 4.14 (Cube roots: $r = 3$) For

$$z_{n+1}^3 = -z_n,$$

the cycle becomes

$$\left\{e^{i\pi/4}, e^{-i\pi/4}\right\},$$

and
$$\alpha_n = \frac{\pi}{4} + \left(-\frac{1}{3}\right)^n \left(\alpha_0 - \frac{\pi}{4}\right).$$
The full inverse level has 3^n vertices, making the transition from binary to ternary branch coding completely explicit.

4.8 Complete periodic classification

Theorem 4.15 (Periodic points of the principal negative-root map). *Let $r \geq 2$.*

(a) *0 is the unique fixed point of F_r.*

(b) *The pair $\{\zeta_+, \zeta_-\}$ is the unique nonzero periodic orbit.*

(c) *There are no nonzero periodic points of period greater than two.*

Proof. Suppose first that z is a periodic point. Then its modulus is periodic under
$$\rho \mapsto \rho^{1/r}.$$
Writing $u = \log \rho$, this becomes
$$u \mapsto \frac{u}{r}.$$
A nonzero real number cannot be periodic under multiplication by $1/r$. Hence either $\rho = 1$ or $z = 0$. Thus every nonzero periodic point lies on the unit circle.

Now consider the signed magnitude $\alpha = |\operatorname{Arg} z|$. It is periodic under the contraction
$$A_r(\alpha) = \frac{\pi - \alpha}{r}.$$
If α has period p, then
$$\alpha = A_r^p(\alpha).$$
But A_r^p is an affine contraction of slope $(-1/r)^p$, hence has a unique fixed point. Since the fixed point of A_r itself is $\pi/(r+1)$, this is the only periodic value of α.

Thus every nonzero periodic point must have angular magnitude $\pi/(r+1)$. The principal map reverses the sign, producing exactly the two-cycle $\{\zeta_+, \zeta_-\}$. □

Remark 4.16 The two-cycle is therefore not an unexplained numerical phenomenon. It is forced by two independent contractions: radial contraction in logarithmic coordinates and angular contraction with orientation reversal.

5 Phase-perturbed principal root dynamics

The fixed multiplier $-1 = e^{i\pi}$ is only one member of a broader family. For $\beta \in \mathbb{R}$, define
$$F_{r,\beta}(z) = \operatorname{Root}_r^{\mathrm{pr}}(e^{i\beta} z).$$
The modulus remains completely rigid:
$$|F_{r,\beta}(z)| = |z|^{1/r}.$$
Thus the phase parameter changes only the angular dynamics. This section makes that angular system explicit while being careful about a subtle point: taking a principal rth root is not the same operation as multiplying an angle by $1/r$ on the circle. The former requires a branch choice for the covering map $w \mapsto w^r$, and the principal-value convention records that choice through an integer-valued reduction function.

5.1 Principal-angle form and reduction index

For $\theta \in (-\pi, \pi]$, set

$$\kappa_\beta(\theta) \in \mathbb{Z}$$

to be the unique integer satisfying

$$-\pi < \theta + \beta - 2\pi\kappa_\beta(\theta) \le \pi.$$

Then

$$\mathrm{PV}(\theta + \beta) = \theta + \beta - 2\pi\kappa_\beta(\theta),$$

and the principal-root angle map is exactly

$$T_{r,\beta}(\theta) = \frac{\theta + \beta - 2\pi\kappa_\beta(\theta)}{r}. \tag{10}$$

The function κ_β is piecewise constant, with jumps precisely at those values for which $\theta + \beta$ crosses a principal-argument boundary. Away from such boundaries, $T_{r,\beta}$ is affine with slope $1/r$. The map is therefore a piecewise-affine contraction on the principal angle interval.

Remark 5.1 (Why the circle model must retain the branch index) The quotient $\theta \bmod 2\pi$ forgets which representative is used before the root is extracted. Since division by r is not well-defined on $\mathbb{R}/(2\pi\mathbb{Z})$ without selecting a lift, the principal root naturally lives on a chosen angular interval rather than as a globally well-defined affine endomorphism of the circle. This is precisely the mathematical content of the branch-selection rule.

5.2 Relation with contracted rotations

The systems

$$x \longmapsto \lambda x + \delta \pmod 1, \qquad 0 < \lambda < 1,$$

known as contracted rotations or λ-affine circle contractions, form a classical family; see, for example, Laurent–Nogueira [5]. Before the adapted coordinate is introduced, the root map is best regarded as a branch-selected piecewise-affine contraction: the integer $\kappa_\beta(\theta)$ appears before division by r, and the image is confined to the narrow interval $(-\pi/r, \pi/r]$. The important point established below is stronger than a mere analogy: after restriction to this post-root interval and the affine coordinate Ξ_r, the family is exactly the standard contracted rotation $x \mapsto \{x/r + \delta\}$.

For comparison, if $x = (\theta + \pi)/(2\pi)$, then (10) becomes

$$x_{n+1} = \frac{x_n}{r} + \frac{r-1}{2r} + \frac{\beta}{2\pi r} - \frac{\kappa_\beta(\theta_n)}{r}. \tag{11}$$

The branch index in (11) is indispensable. This formula also corrects a tempting but incorrect simplification in which the phase term is written as $\beta/(2\pi)$ without the factor $1/r$.

5.3 Fixed points and endpoint-sensitive phase intervals

A nonzero fixed point must have modulus one. Let $z = e^{i\theta}$ with $\theta \in (-\pi, \pi]$. If the principal-reduction index at the fixed point is k, then

$$r\theta = \theta + \beta - 2\pi k,$$

so the formal candidate is

$$\theta_k = \frac{\beta - 2\pi k}{r - 1}. \tag{12}$$

The candidate is genuine only if two conditions hold simultaneously:

$$-\pi < \theta_k \le \pi, \tag{13}$$
$$-\pi < r\theta_k \le \pi. \tag{14}$$

The second condition is not optional: it is the statement that the quantity $\theta + \beta - 2\pi k = r\theta$ is actually the principal representative selected before taking the root.

For the central branch $k = 0$, condition (14) gives

$$-\pi + \frac{\pi}{r} < \beta \le \pi - \frac{\pi}{r}. \tag{15}$$

If β is itself restricted to the principal phase range $(-\pi, \pi]$, the other integer branches are excluded from this central window. The open left endpoint and closed right endpoint are consequences of the convention $\operatorname{Arg} \in (-\pi, \pi]$, not an arbitrary stylistic choice.

Remark 5.2 (Endpoint discipline) At a boundary parameter, replacing $(-\pi, \pi]$ by a symmetric closed or open interval can change which integer k is assigned to the same geometric angle. Therefore phase bifurcation statements should always specify the argument convention together with the inequality convention.

5.4 Periodic itineraries

The branch reduction occurring at each step can be encoded by an integer symbol. For a prescribed itinerary $\boldsymbol{k} = (k_0, \ldots, k_{p-1})$, define

$$\theta_{n+1} = \frac{\theta_n + \beta - 2\pi k_n}{r}, \qquad n = 0, \ldots, p-1.$$

Iterating gives

$$\theta_p = \frac{\theta_0}{r^p} + \sum_{j=0}^{p-1} \frac{\beta - 2\pi k_j}{r^{p-j}}. \tag{16}$$

Hence a periodic point with $\theta_p = \theta_0$ must satisfy

$$\boxed{\theta_0 = \frac{\sum_{j=0}^{p-1} r^j (\beta - 2\pi k_j)}{r^p - 1}.} \tag{17}$$

Equation (17) gives a candidate, not an automatic orbit. For every j, one must also verify the finite admissibility inequalities

$$-\pi < \theta_j + \beta - 2\pi k_j \le \pi. \tag{18}$$

This is a complete finite test for a prescribed itinerary.

Proposition 5.3 (Finite symbolic admissibility). *For fixed $r \ge 2$, $\beta \in \mathbb{R}$, and a finite word $\boldsymbol{k} = (k_0, \ldots, k_{p-1})$, the candidate in (17) is a genuine periodic orbit of the principal phase map if and only if the recursively generated angles satisfy (18) at every step.*

Proof. If the orbit is genuine, each k_j is by definition the integer that places $\theta_j + \beta - 2\pi k_j$ in the principal interval, so the inequalities are necessary. Conversely, if all inequalities hold, the affine recurrence specified by the word agrees with the principal recurrence at every step. Equation (17) is exactly the condition $\theta_p = \theta_0$, hence the candidate closes to a periodic orbit. □

Remark 5.4 (Primitive words and orbit multiplicity) A word of length p may be a repetition of a shorter word. Such a word produces a periodic point whose primitive period divides p. Also, cyclic rotations of a word generally describe the same orbit with a different starting point. A classification of primitive cycles must therefore quotient by both word repetition and cyclic rotation.

Proposition 5.5 (Finite-word parameter sets are interval-computable). *Fix $r \geq 2$ and a finite word $\boldsymbol{k} = (k_0, \ldots, k_{p-1})$. After substituting the candidate (17) into the admissibility inequalities (18), every condition is an affine inequality in β. Consequently, the set of phases for which this prescribed word is admissible is an intersection of finitely many half-lines in $\mathbb{R}$ and hence is either empty, a point, or an interval (before reduction modulo 2π).*

Proof. The candidate θ_0 in (17) is affine in β. Every subsequent θ_j obtained from the affine recurrence is therefore affine in β. Hence each quantity $\theta_j + \beta - 2\pi k_j$ is affine in β, and each admissibility condition has the form $A_j\beta < B_j$ or $A_j\beta \leq B_j$. A finite intersection of such one-dimensional half-lines is an interval, a singleton, or the empty set. □

Remark 5.6 (From a word to a locking window) This proposition turns the finite symbolic test into an exact parameter-space algorithm: choose a word, compute its unique periodic candidate, substitute into the finite branch inequalities, and intersect the resulting linear constraints. The global rotation number then identifies which such intervals belong to the same rational mode-locked plateau.

5.5 A useful contraction estimate for a fixed itinerary

Whenever two points follow the same admissible branch itinerary for p steps, their principal angles satisfy

$$\theta_p - \widetilde{\theta}_p = \frac{\theta_0 - \widetilde{\theta}_0}{r^p}.$$

Thus the branch itinerary itself carries an exponentially stable affine recurrence. All complexity in the phase-parameter problem therefore comes not from the local affine dynamics but from the combinatorics of which itinerary is admissible as parameters or initial conditions vary.

This separation is useful later: it identifies branch boundaries—not weak contraction—as the natural source of bifurcation phenomena.

6 Exact contracted-rotation representation of the phase family

The piecewise-affine description in the preceding section can be sharpened. The key point is that the principal root does not use the entire principal angle interval as its range. After one application of the root, every angle is confined to the smaller interval

$$I_r := \left(-\frac{\pi}{r}, \frac{\pi}{r}\right].$$

This interval is forward invariant for every value of the phase parameter. Consequently, the correct normalization for comparison with contracted rotations is the affine coordinate on I_r, rather than the usual coordinate on $(-\pi, \pi]$.

6.1 The post-root invariant interval

For every $\theta \in (-\pi, \pi]$ and every $\beta \in \mathbb{R}$, (10) gives

$$-\frac{\pi}{r} < T_{r,\beta}(\theta) \leq \frac{\pi}{r}.$$

Thus I_r contains the image of the entire principal-angle map. In particular, if θ_n is generated by the phase-perturbed root recurrence, then $\theta_n \in I_r$ for every $n \geq 1$.

Lemma 6.1 (Forward invariance of the root-angle interval). *For every $r \geq 2$ and every $\beta \in \mathbb{R}$,*

$$T_{r,\beta}((-\pi, \pi]) \subset I_r.$$

Hence I_r is forward invariant under $T_{r,\beta}$.

Proof. By construction the principal representative

$$\theta + \beta - 2\pi\kappa_\beta(\theta)$$

lies in $(-\pi, \pi]$. Division by r therefore puts the output in $(-\pi/r, \pi/r]$. The same conclusion applies to inputs already in I_r. ☐

The contraction aspect is now particularly transparent. If two points of I_r follow the same local branch, their angular difference is divided by r. The branch discontinuities are encoded by the integer reduction index, while the continuous part of the dynamics is a uniform contraction.

6.2 The natural contracted-rotation coordinate

For $\theta \in I_r$, define

$$x = \Xi_r(\theta) := \frac{\pi - r\theta}{2\pi}. \tag{19}$$

Because $\theta \in (-\pi/r, \pi/r]$, this gives

$$\Xi_r : I_r \longrightarrow [0, 1),$$

and it is an affine bijection with inverse

$$\theta = \Xi_r^{-1}(x) = \frac{\pi - 2\pi x}{r}.$$

Theorem 6.2 (Exact contracted-rotation conjugacy). *Fix $r \geq 2$ and $\beta \in \mathbb{R}$. For every $n \geq 1$, let $\theta_n = \operatorname{Arg} z_n$ for the principal phase-perturbed recurrence*

$$z_{n+1} = F_{r,\beta}(z_n).$$

Define

$$x_n = \Xi_r(\theta_n).$$

Then

$$x_{n+1} = \left\{ \frac{x_n}{r} + \delta_{r,\beta} \right\}, \tag{20}$$

where

$$\delta_{r,\beta} := \frac{r-1}{2r} - \frac{\beta}{2\pi} \pmod 1. \tag{21}$$

Thus, after at most the first root extraction, the angular dynamics is exactly a standard contracted rotation with contraction ratio $1/r$.

Proof. For $n \geq 1$, write

$$\theta_{n+1} = \frac{\theta_n + \beta - 2\pi k_n}{r}$$

for the unique principal reduction index k_n. Then

$$\begin{aligned} x_{n+1} &= \frac{\pi - r\theta_{n+1}}{2\pi} \\ &= \frac{\pi - (\theta_n + \beta - 2\pi k_n)}{2\pi} \\ &= \frac{1}{2} - \frac{\theta_n}{2\pi} - \frac{\beta}{2\pi} + k_n. \end{aligned}$$

Using

$$\theta_n = \frac{\pi - 2\pi x_n}{r}$$

from the inverse coordinate relation gives

$$x_{n+1} = \frac{x_n}{r} + \frac{r-1}{2r} - \frac{\beta}{2\pi} + k_n.$$

Reduction modulo 1 removes the integer k_n, yielding (20). □

Definition 6.3 (Fractional-part convention) For $y \in \mathbb{R}$, write

$$\{y\} := y - \lfloor y \rfloor \in [0, 1).$$

This is the convention used throughout the contracted-rotation formulation.

Remark 6.4 (Why this does not contradict the branch-sensitive formulation) The earlier warning about division of angles modulo 2π remains correct. The exact contracted-rotation structure appears only after the range of the principal root has been used to select its natural affine coordinate. The coordinate Ξ_r is not the naive quotient coordinate $\theta \mapsto \theta/(2\pi)$; it is adapted to the inverse-covering geometry of the root map. In this coordinate the branch reduction integer becomes exactly the integer removed by the fractional-part operation.

6.3 The original negative-root system as a distinguished parameter

For the original recurrence $z_{n+1}^r = -z_n$, one has $\beta = \pi$. Hence

$$\delta_{r,\pi} = -\frac{1}{2r} \pmod 1,$$

and the normalized phase dynamics is

$$x_{n+1} = \left\{ \frac{x_n}{r} - \frac{1}{2r} \right\}. \tag{22}$$

The two-cycle from the principal-root analysis has normalized coordinates

$$x_- = \frac{1}{2(r+1)}, \qquad x_+ = \frac{2r+1}{2(r+1)},$$

which satisfy

$$x_- = C_{r,\pi}(x_+), \qquad x_+ = C_{r,\pi}(x_-),$$

where $C_{r,\pi}$ denotes the contracted rotation in (22). Thus the principal two-cycle is a rotation-number-1/2 phenomenon in the contracted-rotation representation.

Proposition 6.5 (Two-cycle as rotation-number-1/2 dynamics). *For every $r \geq 2$, the contracted rotation associated with $\beta = \pi$ has rotation number $1/2$ in the standard contracted-rotation sense. Its attracting period-two orbit is the image under Ξ_r^{-1} of the pair x_-, x_+.*

Proof. The displayed points form a genuine two-cycle. Directly lifting the contracted rotation along this period-two orbit gives a net translation of one unit over two iterates, hence rotation number $1/2$. Uniqueness of the corresponding attracting orbit also follows from the exact radial and angular contractions proved earlier. □

6.4 Rotation-number parameterization

Let $\rho(\lambda, \delta)$ denote the rotation number in the standard theory of the contracted rotation

$$C_{\lambda,\delta}(x) = \{\lambda x + \delta\}, \qquad 0 < \lambda < 1,$$

with $\delta \in [0, 1)$. Laurent and Nogueira develop the dependence of this rotation number on λ and δ, including its arithmetic relation to Hecke–Mahler series [5]. In the present family,

$$\lambda = \frac{1}{r}, \qquad \delta = \delta_{r,\beta}.$$

It is therefore natural to define the phase rotation number by

$$\rho_r(\beta) := \rho\left(\frac{1}{r}, \delta_{r,\beta}\right). \tag{23}$$

This function is 2π-periodic in β because only $e^{i\beta}$ enters the original complex map.

The value of (23) is not merely a numerical summary. It organizes the global asymptotic phase behavior. Rational values correspond to periodic symbolic regimes, while irrational values correspond to the non-periodic contracted-rotation regime and its invariant Cantor-type attractor. The detailed statements below are imported from the established contracted-rotation theory after the exact conjugacy theorem.

Theorem 6.6 (Phase arithmetic inherited from contracted rotations). *Fix $r \geq 2$.*

(a) *If $\delta_{r,\beta}$ is algebraic, then $\rho_r(\beta)$ is rational.*

(b) *If $\rho_r(\beta)$ is irrational, then β/π is transcendental.*

(c) *The set of phases modulo 2π for which $\rho_r(\beta)$ is irrational has Hausdorff dimension zero.*

Proof. Part (a) is the algebraicity-to-rationality theorem for contracted rotations proved by Laurent and Nogueira, applied with $\lambda = 1/r$ and $\delta = \delta_{r,\beta}$ [5]. For (b), if β/π were algebraic, then $\delta_{r,\beta}$ would be algebraic because $(r-1)/(2r)$ is rational. Part (a) would then force $\rho_r(\beta)$ to be rational, a contradiction. Part (c) follows from the zero-Hausdorff-dimension result for the irrational/exceptional parameter set of contracted rotations and, more generally, from the piecewise-affine contraction result of Gaivão [7]. □

Remark 6.7 (One-way arithmetic implication) The transcendence conclusion in part (b) is intentionally one-way. A transcendental phase may still produce a rational rotation number, because rational rotation numbers can occupy nontrivial parameter intervals in the contracted-rotation family. The theorem does not assert the converse.

6.5 Rational plateaus and mode locking

For fixed contraction ratio, rational rotation numbers occur on parameter intervals in the contracted-rotation family. Their pullbacks under $\beta \mapsto \delta_{r,\beta}$ therefore produce phase intervals with common asymptotic periodic combinatorics. This is the root-theoretic form of mode locking: a finite symbolic regime can persist under a nontrivial range of phase perturbations.

Proposition 6.8 (Pullback of rational parameter intervals). *Suppose that for fixed $\lambda = 1/r$ the contracted rotation has a parameter interval $[\delta_-, \delta_+]$ on which the rotation number equals p/q. Then the corresponding phase set is the interval obtained from*

$$\delta_{r,\beta} = \frac{r-1}{2r} - \frac{\beta}{2\pi}$$

by affine inversion, taken modulo 2π. On this phase interval the root system has rotation number p/q and asymptotic periodic combinatorics of period q.

Proof. The map $\beta \mapsto \delta_{r,\beta}$ is affine with nonzero slope. Thus its preimage of a parameter interval is an interval, and the exact conjugacy transfers the rotation number and its periodic classification. □

This refines the finite-word description of Section 5. The admissibility inequalities there test individual symbolic words. The rotation number instead organizes all admissible periodic words into global parameter regions.

6.6 Explicit period-one and period-two mode-locking windows

The affine structure permits two particularly transparent locking intervals. They are useful because they anchor the abstract rotation-number discussion in the original complex-root parameter β.

Proposition 6.9 (Central period-one plateau). *For the contracted rotation*

$$C_{r,\delta}(x) = \left\{ \frac{x}{r} + \delta \right\},$$

the zero-rotation fixed point with branch index 0 exists whenever

$$0 \le \delta < \frac{r-1}{r}.$$

Equivalently, for phases represented modulo 2π and chosen near $\beta = 0$, the corresponding principal-root system has a fixed point on the central branch throughout

$$-\pi + \frac{\pi}{r} < \beta \le \pi - \frac{\pi}{r}.$$

The fixed point is

$$x_* = \frac{r\delta}{r-1}, \qquad \theta_* = \frac{\beta}{r-1}$$

on the branch for which $k = 0$.

Proof. For a fixed point with no fractional-part wrap one has $x = x/r + \delta$, hence $x = r\delta/(r-1)$. The condition $x \in [0,1)$ is exactly $0 \le \delta < (r-1)/r$. Substitution of $\delta = (r-1)/(2r) - \beta/(2\pi)$ gives the stated phase interval. In root coordinates the fixed-point equation with $k = 0$ is $r\theta = \theta + \beta$, hence $\theta = \beta/(r-1)$. □

The next window is more revealing for the original negative-root recurrence.

Theorem 6.10 (Explicit period-two mode-locking window). *For the* 01 *itinerary of the contracted rotation with $\lambda = 1/r$, the periodic point is*

$$x_0 = \frac{r((r+1)\delta - r)}{r^2 - 1}, \tag{24}$$

$$x_1 = \frac{r((r+1)\delta - 1)}{r^2 - 1}. \tag{25}$$

The itinerary is admissible on the parameter interval

$$\frac{r}{r+1} \le \delta < 1 - \frac{1}{r(r+1)}. \tag{26}$$

Its interior

$$\frac{r}{r+1} < \delta < 1 - \frac{1}{r(r+1)} \tag{27}$$

is a robust period-two regime: the two branch inequalities are strict, so the period-two symbolic itinerary persists under small parameter perturbations. The midpoint of this interval is

$$1 - \frac{1}{2r}.$$

Consequently, since $\delta_{r,\pi} = 1 - 1/(2r)$ modulo 1*, the original negative-root system $\beta = \pi$ lies exactly at the midpoint of this period-two window.*

Proof. For the word 01 write

$$x_1 = \frac{x_0}{r} + \delta, \qquad x_0 = \frac{x_1}{r} + \delta - 1.$$

Solving gives the displayed formulas for x_0 and x_1. Admissibility requires $0 \le x_0 < 1$ and $0 \le x_1 < 1$. The strongest lower bound is $x_0 \ge 0$, namely $\delta \ge r/(r+1)$, while the strongest upper bound is $x_1 < 1$, namely $\delta < 1 - 1/[r(r+1)]$. The remaining inequalities are weaker for $r \ge 2$. Strictness of both interior inequalities is immediate. Averaging the two endpoints gives $1 - 1/(2r)$. For $\beta = \pi$,

$$\delta_{r,\pi} = \frac{r-1}{2r} - \frac{1}{2} = -\frac{1}{2r} \pmod 1 = 1 - \frac{1}{2r},$$

which is the midpoint. □

Corollary 6.11 (Phase form of the period-two window). *Viewed modulo 2π, the robust period-two interval in* (27) *is equivalent to*

$$|\beta - \pi| < \frac{\pi(r-1)}{r(r+1)} \tag{28}$$

with distance understood modulo 2π. In particular, the original negative root case is not an isolated period-two parameter: it is centered in an explicit open phase neighborhood having the same period-two symbolic regime.

Proof. The map $\beta \mapsto \delta_{r,\beta}$ has slope $-1/(2\pi)$. The interval (27) has half-width $(r-1)/(2r(r+1))$ around $1 - 1/(2r)$. Multiplying by 2π gives (28). □

Remark 6.12 (Endpoint behavior) At the endpoints of (26), one branch orbit lands on the fractional-part boundary. The principal convention $[0, 1)$ then assigns a preferred representative, but the symbolic word is no longer robust under perturbation. For this reason the open interval (27) is the clean statement of structural stability, while the half-open interval records the exact finite-word admissibility under the chosen endpoint convention.

6.7 Irrational phases and Cantor attractors

For an irrational rotation number, the contracted-rotation theory gives a Cantor attractor on which the dynamics is topologically conjugate to an irrational rigid rotation. This explains how a system with contracting affine branches can possess a non-periodic asymptotic set: the contraction acts within branches, while discontinuity and branch switching preserve symbolic complexity.

Theorem 6.13 (Irrational phase regime). *Suppose $\rho_r(\beta)$ is irrational. Then the post-first-iterate angular attractor of the principal phase map is a Cantor set in I_r, and the restriction of the phase map to that attractor is topologically conjugate to an irrational circle rotation. In particular, no point of the attractor is periodic.*

Proof. The exact conjugacy identifies the post-first-iterate phase map with the contracted rotation having $\lambda = 1/r$ and $\delta = \delta_{r,\beta}$. The asserted Cantor and irrational-rotation classification is the standard irrational regime of contracted rotations [5]. Conjugacy transfers the statement back to the root coordinates. □

Remark 6.14 (Recent piecewise-contraction context) The exact root family is a particularly simple one-break piecewise-affine contraction. Recent results for substantially broader classes prove that the exceptional set of parameters producing non-asymptotically-periodic behavior has Hausdorff dimension zero [7, 6]. The root model therefore supplies an explicit complex-analytic realization of a phenomenon that is part of an active one-dimensional dynamical theory.

6.8 The global phase attractor in the complex plane

The exact conjugacy does more than classify angular symbolic behavior. It permits the entire nonzero complex asymptotic set of the phase-perturbed root recursion to be written directly in terms of the contracted-rotation attractor.

Let $K_{r,\beta} \subset [0,1)$ denote the asymptotic attractor of $C_{r,\delta_{r,\beta}}$.

Theorem 6.15 (Global phase-attractor reduction). *For $z_0 \neq 0$, let (z_n) satisfy*

$$z_{n+1} = F_{r,\beta}(z_n).$$

Then

$$\omega(z_0) = \left\{ \exp\Big(i\,\Xi_r^{-1}(x)\Big) : x \in K_{r,\beta} \right\}. \tag{29}$$

In particular, the nonzero complex ω-limit set is independent of $|z_0|$. More precisely:

(a) *if $\rho_r(\beta)$ is rational, the asymptotic set is a finite periodic orbit on $\mathbb{S}^1$;*

(b) *if $\rho_r(\beta)$ is irrational, the asymptotic set is a Cantor subset of $\mathbb{S}^1$, and the restricted dynamics is topologically conjugate to an irrational rigid rotation;*

(c) *in every case the radial coordinate contributes no additional asymptotic freedom, because $\log|z_n| = (\log|z_0|)/r^n \to 0$.*

Proof. For $n \geq 1$, the angle θ_n belongs to I_r and satisfies $x_n = \Xi_r(\theta_n)$. By the exact conjugacy, $x_{n+1} = C_{r,\delta_{r,\beta}}(x_n)$. Hence the accumulation set of the angles is precisely $\Xi_r^{-1}(K_{r,\beta})$. At the same time,

$$z_n = \exp\left(\frac{\log|z_0|}{r^n}\right)\exp(i\theta_n),$$

and the first factor converges to 1. Therefore multiplication by the radial factor does not alter the accumulation set, which proves (29). The rational and irrational cases are the standard alternatives supplied by the contracted-rotation theory cited above. □

Remark 6.16 (A single organizing theorem) The theorem places the deterministic negative-root cycle, the phase-perturbed periodic plateaus, and the irrational Cantor regime on one logical chain: radial contraction reduces the complex problem to its unit-circle angle, the principal root compresses that angle into I_r, and the affine coordinate Ξ_r identifies the remaining dynamics exactly with a contracted rotation. No additional asymptotic mechanism is required.

6.9 A sharper arithmetic consequence at reciprocal-integer contraction

When $\lambda = 1/r$, Laurent–Nogueira's analysis ties the irrational parameter to Hecke–Mahler series and continued-fraction data. In the reciprocal-integer case, the associated parameter can be characterized as Liouville precisely in the regime in which the partial quotients of the irrational rotation number are unbounded [5]. The affine relation between δ and β/π transfers this property to the phase parameter.

Theorem 6.17 (Liouville refinement). *Assume $\rho_r(\beta)$ is irrational and write*

$$\rho_r(\beta) = [0; a_1, a_2, a_3, \ldots]$$

for its continued fraction. In the reciprocal-integer contraction regime $\lambda = 1/r$, the corresponding phase parameter β/π is Liouville if and only if the sequence (a_n) is unbounded, under the standard identification of the contracted-rotation parameter with the phase parameter.

Proof. Laurent and Nogueira establish the stated Liouville criterion for the reciprocal-integer contraction parameter [5]. The phase parameter is related by the nonzero rational affine transformation

$$\frac{\beta}{\pi} = (r-1) - 2r\delta_{r,\beta}.$$

Nonzero rational affine transformations preserve the Liouville property, which gives the result. □

Remark 6.18 This arithmetic theorem is not needed for the solution of the original negative-root map. Its role is to show that the phase extension moves the problem into a genuinely arithmetic part of one-dimensional dynamics. The exact branch normalization is what makes that transfer transparent.

6.10 Endpoint conventions revisited

The endpoint convention $(-\pi, \pi]$ remains important after passing to $[0, 1)$. Boundary points have a preferred representative, and the corresponding symbolic word may depend on this half-open convention even though the underlying geometric phase orbit does not. Consequently, any statement about a particular finite itinerary should retain the explicit admissibility inequalities of Section 5, even when a global rotation-number theorem is invoked.

7 The full multivalued correspondence and inverse dynamics

The full negative-root correspondence is most naturally expressed as the inverse relation of a polynomial.

Definition 7.1 Define

$$P_r : \mathbb{C} \longrightarrow \mathbb{C}, \qquad P_r(z) = -z^r.$$

Then

$$\mathcal{F}_r = P_r^{-1}$$

as a set-valued inverse correspondence.

Thus repeated application of all roots is precisely inverse iteration of P_r.

For $z_0 \in \mathbb{C}^\times$, define the depth-$n$ inverse set

$$A_n(z_0) = P_r^{-n}(\{z_0\}).$$

7.1 Exact iterates of the monomial

Lemma 7.2 (Exact formula for P_r^n). *For $n \geq 1$,*

$$P_r^n(w) = (-1)^{1+r+r^2+\cdots+r^{n-1}} w^{r^n}. \tag{30}$$

Proof. The assertion is true for $n = 1$. Suppose it holds at depth n. Then

$$P_r^{n+1}(w) = P_r(P_r^n(w)) = -\left((-1)^{1+r+\cdots+r^{n-1}} w^{r^n}\right)^r,$$

which equals

$$(-1)^{1+r+\cdots+r^n} w^{r^{n+1}}.$$

$\square$

Consequently,

$$A_n(z_0) = \left\{w \in \mathbb{C} : w^{r^n} = (-1)^{1+r+\cdots+r^{n-1}} z_0\right\}. \tag{31}$$

7.2 The sign exponent and parity simplification

The exponent

$$S_n := 1 + r + r^2 + \cdots + r^{n-1} = \frac{r^n - 1}{r - 1}$$

in the iterate formula is itself elementary but useful. Its parity gives a particularly simple description of the orientation of the inverse polygon.

Proposition 7.3 (Parity of the inverse sign). *For $r \geq 2$,*

$$(-1)^{S_n} = \begin{cases} -1, & r \text{ even}, \\ (-1)^n, & r \text{ odd}. \end{cases}$$

Hence, for even r, every inverse level is a regular r^n-gon obtained from the positive-root polygon by a fixed half-turn at the terminal equation, whereas for odd r the sign alternates with depth.

Proof. If r is even, every term r^j for $j \geq 1$ is even, so $S_n \equiv 1 \pmod 2$. If r is odd, every term is odd, so $S_n \equiv n \pmod 2$. $\square$

Thus one may write

$$A_n(z_0) = \left\{\rho_n \exp\left(i\frac{\phi_n + 2\pi j}{r^n}\right) : j = 0, \ldots, r^n - 1\right\}, \tag{32}$$

where

$$\rho_n = |z_0|^{1/r^n}, \qquad \phi_n = \operatorname{Arg}\left((-1)^{S_n} z_0\right).$$

Equation (32) is the canonical terminal-level parametrization used in the geometric arguments below.

7.3 Cardinality and polygon geometry

Theorem 7.4 (Inverse levels are regular polygons). *Let $z_0 \neq 0$. Then $A_n(z_0)$ contains exactly r^n distinct points. Every point has modulus*

$$\rho_n = |z_0|^{1/r^n},$$

and the arguments are equally spaced with angular separation

$$\frac{2\pi}{r^n}.$$

Thus $A_n(z_0)$ is the vertex set of a rotated regular r^n-gon centered at the origin.

Proof. The equation (31) is

$$w^{r^n} = c_n, \qquad c_n \neq 0.$$

A nonzero complex number has exactly r^n distinct r^nth roots. Writing

$$c_n = |c_n|\, e^{i\phi_n},$$

the roots are

$$\rho_n \exp\left(i\frac{\phi_n + 2\pi j}{r^n}\right), \qquad j = 0, \ldots, r^n - 1.$$

□

Remark 7.5 The inverse tree of a general polynomial can be geometrically complicated. The monomial form here is exceptional: the inverse equation collapses to one explicit algebraic equation of degree r^n. This is the basic reason the present family is exactly solvable at arbitrary depth.

8 Branch words, base-r coding, and rooted trees

The inverse level contains r^n points, while a depth-n branch history contains n choices from an alphabet of size r. The equality $r^n = r \cdot \ldots \cdot r$ therefore suggests symbolic coding. There is, however, a useful distinction between a *chronological branch history* and a *terminal root index*. Keeping that distinction visible prevents a purely notational identification from being mistaken for a theorem about the order in which branches are selected.

8.1 Terminal root indices and base-r words

The explicit inverse equation gives a terminal index

$$j \in \{0, \ldots, r^n - 1\}.$$

Every such integer has a unique length-n base-r expansion

$$j = k_1 r^{n-1} + k_2 r^{n-2} + \cdots + k_n, \qquad k_n \in \{0, \ldots, r-1\}. \tag{33}$$

Proposition 8.1 (Base-r terminal coding). *The map*

$$(k_1, \ldots, k_n) \longmapsto k_1 r^{n-1} + \cdots + k_n$$

is a bijection from $\{0, \ldots, r-1\}^n$ onto $\{0, \ldots, r^n - 1\}$. Consequently the terminal inverse level admits a canonical coding by length-n base-r words once the ordering in (32) has been fixed.

Proof. This is the uniqueness theorem for finite base-r expansions. □

8.2 Chronological histories versus terminal labels

A chronological branch history records which of the r roots was chosen at each successive extraction. Its relation to the terminal index j depends on the precise labeling convention for roots and on the sign rotations introduced by the map $-z^r$. Therefore the strongest convention-independent statement is the following: there is a canonical rooted-tree structure on chronological histories, and there is a canonical terminal-level base-r coding after an ordering of the roots has been chosen. These structures are naturally isomorphic, but an explicit digit-by-digit formula requires a declared branch labeling convention.

Proposition 8.2 (Branch histories and inverse levels). *For $z_0 \neq 0$, the set of chronological depth-n histories has cardinality r^n, and the map sending a history to its resulting inverse point is a bijection. After fixing any deterministic labeling of the r roots at every node, this bijection identifies the history set with the vertex set of $A_n(z_0)$. Different labeling conventions differ only by permutations of the vertices at each level.*

Proof. At every nonzero node the equation $w^r = -z$ has exactly r distinct solutions. Thus the history tree has r^n leaves at depth n. If two different histories produced the same point, applying the forward map $P_r = -z^r$ repeatedly would force the two histories to merge at the first level where their branch labels differ, contradicting the distinctness of the roots of a nonzero complex number. Hence the map is injective, and cardinality gives surjectivity onto $A_n(z_0)$. □

Remark 8.3 (Why the distinction matters) The geometric and measure-theoretic results below depend only on the set of vertices and, in the weighted case, on the chosen symbolic measure. They do not depend on whether the digits are read chronologically, reversely, or after a fixed relabeling. Making this explicit prevents an accidental identification of a convenient coordinate convention with an intrinsic property of the correspondence.

8.3 The rooted-tree structure

The histories of all finite lengths form the rooted regular r-ary tree

$$\mathcal{T}_r = \bigsqcup_{n \geq 0} \{0, \dots, r-1\}^n.$$

The root is the empty word, and

$$(k_1, \dots, k_n)$$

has children

$$(k_1, \dots, k_n, j), \qquad j = 0, \dots, r-1.$$

Away from the critical point 0, this is exactly the combinatorial inverse tree of P_r. The exceptional role of 0 is discussed separately below: at 0, all r algebraic roots collapse to the single point 0, so the regular-branching statement is intentionally restricted to $\mathbb{C}^\times$.

8.4 Linear conjugacy with the power map

The sign in

$$P_r(z) = -z^r$$

does not produce a genuinely different global inverse-tree type from the power map.

Proposition 8.4 (Linear conjugacy). *Choose $a \in \mathbb{C}^\times$ satisfying*

$$a^{r-1} = -1.$$

Let $h(z) = az$. Then

$$h^{-1} \circ (z^r) \circ h = P_r.$$

Proof.

$$h^{-1}((h(z))^r) = a^{-1}(az)^r = a^{r-1}z^r = -z^r.$$

□

Thus the inverse-tree structure is linearly conjugate to that of the power map z^r.

8.5 Monodromy interpretation

The iterated monodromy theory of power maps is classical; see Nekrashevych [10]. In that framework, loops around the omitted critical value act on the rooted preimage tree. For the power map, a standard choice of generators yields the familiar r-adic adding-machine behavior. Our use of this language is deliberately interpretative rather than a claim to prove a new theorem about iterated monodromy groups.

The structural correspondence may be summarized as

$$\boxed{\begin{array}{c} \text{root branch history} \longleftrightarrow \text{rooted-tree vertex} \\[4pt] \text{base-}r \text{ terminal code} \longleftrightarrow \text{monodromy action under a chosen labeling.} \end{array}}$$

9 Geometric limits of inverse levels

The inverse polygon $A_n(z_0)$ has radius

$$\rho_n = |z_0|^{1/r^n}.$$

Hence

$$\rho_n \longrightarrow 1.$$

At the same time, its angular mesh

$$\frac{2\pi}{r^n}$$

tends to zero.

This suggests convergence to the unit circle

$$\mathbb{S}^1 = \{z \in \mathbb{C} : |z| = 1\}.$$

Theorem 9.1 (Hausdorff convergence). *For every $z_0 \in \mathbb{C}^\times$,*

$$A_n(z_0) \longrightarrow \mathbb{S}^1$$

in Hausdorff distance.

Proof. Let $\rho_n = |z_0|^{1/r^n}$. Every vertex of A_n has distance

$$|\rho_n - 1|$$

from $\mathbb{S}^1$.

Conversely, the vertices are equally spaced. Every point of $\mathbb{S}^1$ lies within angular distance at most

$$\frac{\pi}{r^n}$$

of some vertex direction. The distance between a point of radius 1 and the corresponding vertex of radius ρ_n is therefore at most

$$\sqrt{\rho_n^2 + 1 - 2\rho_n \cos(\pi/r^n)}.$$

Both quantities tend to zero. □

The preceding argument gives an exact expression.

Theorem 9.2 (Exact Hausdorff distance). *For every $n \geq 1$,*

$$\mathrm{d_H}(A_n, \mathbb{S}^1) = \max\left\{|\rho_n - 1|\,, \sqrt{\rho_n^2 + 1 - 2\rho_n \cos(\pi/r^n)}\right\}. \tag{34}$$

Proof. The first term is exactly the maximal distance from a polygon vertex to the unit circle.

For the reverse directed distance, symmetry reduces the problem to an arc between two consecutive vertex directions. The farthest point of the unit circle from the two adjacent vertices occurs at the midpoint of the arc. The angular separation from the nearest vertex is then π/r^n, giving the second term. □

9.1 Sharp asymptotics

Theorem 9.3 (Sharp geometric rate). *For $z_0 \in \mathbb{C}^\times$,*

$$r^n\,\mathrm{d_H}(A_n, \mathbb{S}^1) \longrightarrow \sqrt{(\log|z_0|)^2 + \pi^2}. \tag{35}$$

Proof. Write

$$L = \log|z_0|\,.$$

Then

$$\rho_n = e^{L/r^n} = 1 + \frac{L}{r^n} + O(r^{-2n}).$$

Hence

$$|\rho_n - 1| = \frac{|L|}{r^n} + O(r^{-2n}).$$

Also,

$$1 - \cos(\pi/r^n) = \frac{\pi^2}{2r^{2n}} + O(r^{-4n}).$$

Therefore

$$\rho_n^2 + 1 - 2\rho_n\cos(\pi/r^n) = (\rho_n - 1)^2 + 2\rho_n(1 - \cos(\pi/r^n)),$$

so

$$= \frac{L^2 + \pi^2}{r^{2n}} + O(r^{-3n}).$$

Taking square roots gives

$$\frac{\sqrt{L^2 + \pi^2}}{r^n} + o(r^{-n}).$$

This second term dominates or equals the radial term asymptotically, and (35) follows. □

Remark 9.4 The limiting constant combines two independent effects: $\log|z|_0$ measures radial displacement, while π is the half-mesh constant of the regular r^n-gon. The Euclidean Hausdorff asymptotic therefore combines radial and angular errors quadratically.

10 Potential theory and the Julia-set interpretation

The exact polygonal formulas obtained above admit a second interpretation in complex dynamics. The polynomial

$$P_r(z) = -z^r$$

is linearly conjugate to the power map $z \mapsto z^r$. Consequently the inverse-level calculations can be viewed as an exactly solvable instance of Julia-set geometry, Green functions, logarithmic capacity, and equilibrium measures.

10.1 Filled Julia set and Julia set

For a polynomial, the filled Julia set is the set of points whose forward orbits remain bounded, and the Julia set is its boundary.

Theorem 10.1 (Julia and filled Julia sets). *For $P_r(z) = -z^r$,*

$$K(P_r) = \overline{\mathbb{D}}, \qquad J(P_r) = \mathbb{S}^1.$$

Moreover, $\mathbb{S}^1$ is completely invariant under P_r.

Proof. If $|z| < 1$, then

$$|P_r^n(z)| = |z|^{r^n} \to 0,$$

so the orbit is bounded. If $|z| > 1$, then

$$|P_r^n(z)| = |z|^{r^n} \to \infty.$$

If $|z| = 1$, every iterate has modulus one. Hence the filled Julia set is the closed unit disk and its boundary is S^1. Complete invariance follows from $|P_r(z)| = 1$ if and only if $|z| = 1$. □

10.2 Green function and the logarithmic radial coordinate

The escape-rate function is explicitly

$$G_r(z) = \max\{0, \log|z|\}. \tag{36}$$

It satisfies

$$G_r(P_r(z)) = rG_r(z). \tag{37}$$

Thus the logarithmic radial coordinate already used throughout the paper is, outside the filled Julia set, precisely the Green-function coordinate.

Proposition 10.2 (Green-function interpretation of the inverse radius). *Let $z_0 \neq 0$ and $w \in A_n(z_0)$. Then*

$$\log|w| = \frac{\log|z_0|}{r^n}.$$

If $|z_0| \geq 1$, this is equivalently

$$G_r(w) = \frac{G_r(z_0)}{r^n}.$$

Proof. The inverse equation gives $|w| = |z_0|^{1/r^n}$, and taking logarithms gives the first identity. The second is its Green-function form outside the unit disk. □

In particular, the fact that inverse levels approach the unit circle can be interpreted as the decay of their Green-function height to zero.

10.3 Equilibrium measure and the Brolin principle

Brolin's theorem identifies the weak limit of normalized inverse-preimage counting measures, under the usual non-exceptionality hypothesis, with the equilibrium measure of the Julia set [8]. In the monomial model this abstract theorem becomes completely explicit: the inverse levels are regular polygons and the equilibrium measure is Haar probability on S^1.

This comparison is useful methodologically. The general theorem establishes universality of inverse-preimage measures, while the present model provides the finite-depth locations, exact Hausdorff error, exact Fourier cancellation, and sharp r^{-n} scaling constant.

Theorem 10.3 (Equilibrium and maximal-entropy interpretation). *The Haar probability m on S^1 is the equilibrium measure of $K(P_r) = \overline{\mathbb{D}}$ and the measure of maximal entropy for $P_r|_{S^1}$. Furthermore,*

$$h_m(P_r) = \log r, \qquad \chi_m(P_r) = \log r.$$

Proof. The map $P_r|_{S^1}$ is conjugate by a rotation to $z \mapsto z^r$. Haar measure is invariant under this map and is the standard maximal-entropy measure, whose metric entropy is $\log r$. Since

$$|P_r'(z)| = r|z|^{r-1} = r$$

on S^1, the Lyapunov exponent is $\log r$. The equilibrium-measure identification is the explicit monomial form of the standard potential- theoretic description. □

Remark 10.4 The purpose of this subsection is not to replace the elementary proof of measure convergence. Rather, it places that proof inside the general theory of polynomial inverse iteration and explains why Haar measure is the canonical limiting distribution in this exactly solvable case.

10.4 Capacity and the geometry of the limiting circle

The logarithmic capacity of the closed unit disk and its boundary is one. This normalization is compatible with (36) and provides a potential-theoretic explanation for the distinguished role of the unit circle. The regular inverse polygons are therefore not merely arbitrary finite sets: they are explicit equidistributed approximations to the equilibrium boundary configuration.

10.5 Quantitative equilibrium approximation

The Wasserstein estimate obtained later in the paper gives

$$W_1(\mu_n, m) = O(r^{-n}).$$

The same scale governs the radial displacement

$$|\log|z_0||\, r^{-n}$$

and the angular half-mesh

$$\pi r^{-n}.$$

The sharp Euclidean constant combines these two errors quadratically. Thus three descriptions of the same approximation coexist: Green-function decay, angular quantization, and Wasserstein convergence.

11 Uniform and weighted branch measures

A finite inverse level can be equipped with a probability measure.

11.1 Uniform branch selection

Define

$$\mu_n = \frac{1}{r^n} \sum_{w \in A_n(z_0)} \delta_w.$$

Since $A_n(z_0)$ is a regular polygon of increasing order and radius approaching one, the limiting measure should be Haar measure m on $\mathbb{S}^1$.

Theorem 11.1 (Uniform branch measure convergence). *For every $z_0 \neq 0$,*

$$\mu_n \Longrightarrow m$$

weakly, where m is normalized Haar measure on the unit circle.[4]

Proof. Let

$$\widehat{\mu_n}(\ell) = \int_{\mathbb{C}} z^\ell \, d\mu_n(z)$$

for integer Fourier modes interpreted on the circle after radial normalization.

The equally spaced r^n-th roots satisfy

$$\frac{1}{r^n} \sum_{j=0}^{r^n-1} e^{2\pi i \ell j / r^n} = 0$$

whenever $r^n \nmid \ell$. Thus for every fixed nonzero integer ℓ, the corresponding Fourier coefficient is exactly zero for all sufficiently large n. The zero coefficient is one.

Since trigonometric polynomials are uniformly dense in continuous functions on $\mathbb{S}^1$, weak convergence to Haar measure follows. The radial convergence $\rho_n \to 1$ transfers the conclusion from the angularly normalized polygons to $A_n(z_0)$ itself. □

Remark 11.2 The convergence is stronger than a generic equidistribution statement: every fixed nonzero Fourier mode vanishes exactly at all sufficiently large finite depths.

11.2 Quantitative weak convergence

The regular-polygon geometry gives more than weak convergence. Let m denote Haar probability on $\mathbb{S}^1$, and let $\widetilde{\mu}_n$ be the uniform probability measure on the angularly normalized polygon obtained from A_n by radial projection onto $\mathbb{S}^1$. Coupling each vertex to the arc determined by its Voronoi cell gives an explicit first-moment estimate.

Proposition 11.3 (A simple Wasserstein bound). *With W_1 denoting the Euclidean Wasserstein-1 distance,*

$$W_1(\mu_n, m) \leq |\rho_n - 1| + 2 \sin\left(\frac{\pi}{r^n}\right).$$

In particular,

$$W_1(\mu_n, m) = O(r^{-n}).$$

[4] Here Haar measure means the unique rotation-invariant Borel probability measure on $\mathbb{S}^1$; it is normalized to have total mass 1.

Proof. Couple a uniformly chosen point of $\mathbb{S}^1$ with the nearest polygon vertex. The angular displacement is at most π/r^n, hence the corresponding chord length at radius one is at most $2\sin(\pi/r^n)$. Moving that point radially from radius one to radius ρ_n costs $|\rho_n - 1|$. Averaging does not increase this uniform pointwise bound. $\square$

11.3 Fourier coefficients at finite depth

The same symmetry gives an exact formula for every integer Fourier mode. If $m \in \mathbb{Z}$, then, writing $c_n = (-1)^{S_n} z_0$ and using the common inverse-level radius ρ_n,

$$\int w^m \, d\mu_n(w) = \begin{cases} \rho_n^m \exp\left(im \dfrac{\operatorname{Arg} c_n}{r^n}\right), & r^n \mid m, \\ 0, & r^n \nmid m. \end{cases} \tag{38}$$

For $m < 0$ this is interpreted in the literal sense $w^m = (1/w)^{|m|}$, which is legitimate because every inverse-level point is nonzero. If one wants purely angular Fourier coefficients, define $\vartheta(w) = w/|w|$ and use $\int \vartheta(w)^m \, d\mu_n(w)$; the radial factor then disappears and the same root-of-unity cancellation remains. Thus every fixed nonzero Fourier mode is eventually exactly zero. This exact cancellation is stronger than merely asserting convergence of Fourier coefficients.

Remark 11.4 (Set convergence versus measure convergence) Hausdorff convergence of supports and weak convergence of probability measures are distinct assertions. The former says that the finite sets become geometrically dense in $\mathbb{S}^1$; the latter says that uniform mass placed on those sets approaches rotationally invariant probability. Neither statement alone implies the other in general, which is why both are recorded separately here.

11.4 Biased branch selection

Let

$$p = (p_0, \ldots, p_{r-1}), \qquad p_k \geq 0, \qquad \sum_{k=0}^{r-1} p_k = 1.$$

Let

$$\Sigma_r = \{0, \ldots, r-1\}^{\mathbb{N}}$$

with product Bernoulli measure

$$\eta_p = \bigotimes_{n=1}^{\infty} p.$$

Define the coding map

$$\Psi((k_n)_{n\geq 1}) = \exp\left(2\pi i \sum_{n=1}^{\infty} \frac{k_n}{r^n}\right). \tag{39}$$

The limiting branch measure is

$$\nu_p = \Psi_{\#}\eta_p.$$

The ambiguity of terminating versus eventually maximal base-r expansions occurs only on a countable set and does not alter the dimension statements below.

Theorem 11.5 (Entropy–dimension relation). *Assume the branch probabilities are independent and identically distributed with probability vector p. Then the measure ν_p is exact dimensional and*

$$\dim_H \nu_p = \frac{h(p)}{\log r}, \tag{40}$$

where

$$h(p) = -\sum_{k=0}^{r-1} p_k \log p_k$$

is the Shannon entropy.

Proof. A cylinder

$$[k_1, \dots, k_n]$$

has probability

$$p_{k_1} \cdots p_{k_n}$$

and geometric diameter comparable to r^{-n} under the coding map. For η_p-almost every sequence, the strong law of large numbers gives

$$-\frac{1}{n} \log(p_{k_1} \cdots p_{k_n}) \longrightarrow h(p).$$

Thus the local dimension in the symbolic metric is

$$\frac{h(p)}{\log r}.$$

Away from the countable endpoint-identification set, the angular coding is locally bi-Lipschitz after identifying the symbolic r-adic interval with its image on the circle. Countable ambiguities do not affect the almost-everywhere dimension. Hence the same dimension is obtained for ν_p. □

Corollary 11.6. *If*

$$p_k = \frac{1}{r}$$

for all k, then

$$\dim_H \nu_p = 1,$$

and ν_p is Haar measure.

Corollary 11.7. *If p is concentrated at a single branch, then*

$$h(p) = 0 \qquad \text{and} \qquad \dim_H \nu_p = 0.$$

Remark 11.8 (Branch information) The phrase "information in branch choices" has a literal mathematical meaning here. At depth n, a branch history contains approximately

$$nh(p)$$

nats of typical information, while its geometric scale is approximately

$$r^{-n}.$$

Their quotient gives the dimension formula

$$\frac{h(p)}{\log r}.$$

11.5 Transfer operators and exact Fourier cancellation

The Fourier cancellation of the uniform inverse measures can be expressed by a normalized transfer operator. On S^1, define

$$(\mathcal{L}f)(z) = \frac{1}{r} \sum_{w \in P_r^{-1}(z)} f(w). \tag{41}$$

For the Fourier character $\chi_m(z) = z^m$ one obtains, after absorbing the fixed sign rotation into a unimodular factor,

$$\mathcal{L}\chi_m = \begin{cases} \omega_m \chi_{m/r}, & r \mid m, \\ 0, & r \nmid m, \end{cases} \tag{42}$$

where $|\omega_m| = 1$. Hence for every fixed nonzero integer m,

$$\mathcal{L}^n \chi_m = 0$$

for all sufficiently large n.

Proposition 11.9 (Haar measure as a transfer fixed point). *The Haar probability m satisfies*

$$\mathcal{L}^* m = m.$$

Moreover, repeated transfer annihilates every fixed nonconstant Fourier mode after finitely many steps.

Proof. The inverse branches of the power map are equally spaced rotations, so their average preserves Haar measure. The Fourier statement is exactly (42). □

Remark 11.10 For a generic polynomial one retains asymptotic equidistribution but loses the finite-step Fourier cancellation. The exact algebraic cancellation is therefore one of the unusually rigid features of the monomial model.

11.6 Entropy, Lyapunov exponents, and dimension

The biased branch measures admit a dynamical reformulation of the entropy– dimension relation. Under the base-r coding, the forward map corresponds to the left shift on the symbolic space, while the derivative on S^1 has constant modulus r.

Theorem 11.11 (Entropy–Lyapunov realization). *Choose $a \in \mathbb{C}$ with $a^{r-1} = -1$ and define the linear conjugacy*

$$h(z) = az, \qquad P_r = h^{-1} \circ Q_r \circ h, \qquad Q_r(z) = z^r.$$

For the Bernoulli branch measure $\nu_p = \Psi_{\#}\eta_p$ define the corresponding P_r-invariant measure

$$\widetilde{\nu}_p := (h^{-1})_{\#}\nu_p.$$

Then

$$h_{\widetilde{\nu}_p}(P_r) = h(p), \qquad \chi_{\widetilde{\nu}_p}(P_r) = \log r,$$

and consequently

$$\dim_H \widetilde{\nu}_p = \frac{h_{\widetilde{\nu}_p}(P_r)}{\chi_{\widetilde{\nu}_p}(P_r)} = \frac{h(p)}{\log r}. \tag{43}$$

The symbolic measure ν_p itself remains the natural Bernoulli measure for the conjugate power map Q_r, not generally for P_r.

Proof. The coding map Ψ intertwines the left shift with $Q_r(z) = z^r$ away from the countable address-identification set, which is irrelevant for the measure-theoretic entropy. Therefore ν_p is Q_r-invariant and has entropy $h(p)$. Since $P_r = h^{-1} \circ Q_r \circ h$, the pushforward $\widetilde{\nu}_p$ is P_r-invariant and has the same entropy. Because $|a| = 1$, h is a Euclidean isometry and therefore preserves Hausdorff dimension. On $\mathbb{S}^1$,

$$|P_r'(z)| = r,$$

so the Lyapunov exponent of $\widetilde{\nu}_p$ is $\log r$. The exact-dimensional formula proved earlier then yields (43). □

Remark 11.12 (Why the conjugacy is necessary) Without the conjugating rotation, a Bernoulli coding measure need not be invariant for $P_r(z) = -z^r$. The correct invariant-measure statement is therefore made first for $Q_r(z) = z^r$ and then transported to P_r through h.

11.7 Frequency classes and a multifractal refinement

Let $q = (q_0, \ldots, q_{r-1})$ be a probability vector and let $E(q)$ denote the set of points whose base-r addresses have asymptotic digit frequencies q_k. Then

$$\dim_H E(q) = \frac{-\sum_{k=0}^{r-1} q_k \log q_k}{\log r}. \tag{44}$$

For the Bernoulli measure ν_p with all $p_k > 0$, the local dimension along such a frequency class is governed by

$$d_{\nu_p}(q) = -\frac{\sum_{k=0}^{r-1} q_k \log p_k}{\log r}. \tag{45}$$

Proposition 11.13 (Frequency-level multifractal spectrum). *The frequency classes $E(q)$ form a multifractal decomposition of the biased branch measure. Their Hausdorff dimensions are given by* (44), *while the local dimensions of ν_p on these classes are described by* (45).

Proof. The number of length-n words with empirical frequency close to q grows on the exponential scale like $\exp(nh(q))$, where $h(q) = -\sum q_k \log q_k$. Cylinders have diameter comparable to r^{-n}, which yields (44). Their ν_p-mass is asymptotically $\exp(n \sum q_k \log p_k)$, giving (45). The two-point endpoint ambiguity of base-r expansions is countable and does not alter these almost-everywhere statements. □

Since $h(q) \le \log r$, with equality only for the uniform vector, Haar is the unique full-dimensional measure among the Bernoulli branch laws considered here.

12 Infinite depth and interpretations of infinite radicals

The phrase "an infinite complex radical" is ambiguous. In the present setting at least six distinct interpretations occur.

1. **Fixed-point interpretation.** Solve

$$z = F_r(z).$$

2. **Principal-orbit interpretation.** Study

$$F_r^n(z_0)$$

for fixed z_0.

3. **Periodic-attractor interpretation.** Study the limiting periodic orbit when the principal orbit does not converge to a single point.
4. **Set-valued interpretation.** Study the inverse sets

$$A_n(z_0).$$

5. **Measure-valued interpretation.** Study probability measures on the inverse levels and their weak limits.
6. **Infinite symbolic interpretation.** Study an infinite branch sequence

$$(k_1, k_2, \ldots) \in \Sigma_r.$$

For the negative-root family these interpretations are genuinely different. The principal orbit has an attracting two-cycle, the inverse sets converge geometrically to the whole unit circle, and weighted branch selection can converge to many different measures.

Thus there is no single mathematically canonical object called "the" infinite negative complex radical without additional specification.

13 Chronological random branch dynamics

The symbolic boundary measure and a random chronological choice of roots are closely related but should not be identified without specifying the time ordering. The distinction is useful because a random sample path, a finite-level counting measure, and a stationary law are different objects even when they are built from the same branch alphabet.

13.1 Random affine recursion

Let $K_1, K_2, \ldots$ be iid random variables with

$$\mathrm{Prob}(K_n = k) = p_k, \qquad k = 0, \ldots, r-1.$$

In the standard inverse-coordinate branches

$$\phi_k(x) = \frac{x+k}{r},$$

the random backward process satisfies

$$X_{n+1} = \frac{X_n + K_{n+1}}{r}. \tag{46}$$

Its solution is

$$X_n = \frac{X_0}{r^n} + \sum_{j=1}^{n} \frac{K_j}{r^{n-j+1}}. \tag{47}$$

Reversing the finite word gives the familiar base-r random series

$$X_\infty = \sum_{j=1}^{\infty} \frac{K_j}{r^j}.$$

The reversal matters for chronological interpretation but not for the resulting Bernoulli boundary law.

Proposition 13.1 (Contractive random boundary limit in distribution). *For every initial condition X_0,*

$$\left| X_n - \sum_{j=1}^{n} \frac{K_{n-j+1}}{r^j} \right| = \frac{|X_0|}{r^n} \longrightarrow 0 \tag{48}$$

for every realization of the bounded iid branch variables. Moreover,

$$\sum_{j=1}^{n} \frac{K_{n-j+1}}{r^j}$$

has the same distribution as

$$Y_n := \sum_{j=1}^{n} \frac{K_j}{r^j}.$$

Consequently X_n converges in distribution to

$$Y := \sum_{j=1}^{\infty} \frac{K_j}{r^j},$$

whose law is the Bernoulli pushforward determined by p.

Proof. The identity (48) follows by reversing the finite sum in (47). Since $(K_1, \dots, K_n)$ has the same joint distribution as $(K_n, \dots, K_1)$, the reversed finite sum has the same distribution as Y_n. The series defining Y converges absolutely almost surely because $0 \le K_j \le r - 1$, and

$$|Y - Y_n| \le (r-1) \sum_{j=n+1}^{\infty} r^{-j} = r^{-n} \longrightarrow 0.$$

Thus $Y_n \to Y$ almost surely. Since $X_n - Y_n$ need not converge to zero on the original chronological sample space, the conclusion for X_n is convergence in distribution, not generally almost-sure convergence. □

13.2 Principal orbit, inverse measure, and random orbit are distinct

The deterministic principal branch selects one branch at every step and hence produces the attracting two-cycle of the original negative-root problem. The full inverse correspondence produces r^n points at depth n. Uniform counting on those points converges to Haar. A random branch process instead produces a random sample path whose distribution approaches the associated symbolic law.

Thus the phrase “the infinite radical converges” has no canonical meaning without specifying the object being iterated and the mode of convergence. In the present model one can simultaneously have deterministic periodic attraction, set convergence to a circle, weak convergence of measures, and convergence in distribution of the chronological random recursion. Almost-sure convergence belongs naturally to the reversed symbolic series used to represent the same limiting Bernoulli law.

Proposition 13.2 (Convergence of first two moments). *Let*

$$\mu_K := \sum_{k=0}^{r-1} k p_k, \qquad \sigma_K^2 := \sum_{k=0}^{r-1} (k - \mu_K)^2 p_k.$$

Then

$$\mathbb{E}[X_n] = \frac{X_0}{r^n} + \frac{\mu_K}{r-1}(1 - r^{-n}),$$

and

$$\operatorname{Var}(X_n) = \frac{\sigma_K^2}{r^2 - 1}(1 - r^{-2n}).$$

Hence

$$\mathbb{E}[X_n] \to \frac{\mu_K}{r - 1}, \qquad \operatorname{Var}(X_n) \to \frac{\sigma_K^2}{r^2 - 1}.$$

Proof. Taking expectation in (47) gives a finite geometric sum. Independence makes the variance of the weighted sum equal to the sum of the squared weights times σ_K^2. The displayed limits follow from $r \geq 2$. □

Remark 13.3 (Stationarity versus deterministic invariance) A stationary distribution belongs to a stochastic transition kernel, whereas an invariant measure belongs to a deterministic map. Uniform branch selection makes the two notions meet at Haar measure in the monomial case, but this is a consequence of the particular symmetry rather than a general identity.

14 Variable root order

The fixed-order recurrence admits a natural nonautonomous extension.

Let

$$r_n \geq 2, \qquad z_n = \operatorname{Root}_{r_n}^{\mathrm{pr}}(-z_{n-1}).$$

Define

$$N_n = \prod_{j=1}^{n} r_j.$$

Proposition 14.1 (Variable-order radial law). *For every n,*

$$|z_n| = |z_0|^{1/N_n}. \tag{49}$$

Proof. Each step raises the modulus to the power $1/r_n$, hence the exponent after n steps is

$$\frac{1}{r_1 r_2 \cdots r_n}.$$

□

Since $r_j \geq 2$,

$$N_n \geq 2^n,$$

so every nonzero variable-order orbit still satisfies

$$|z_n| \to 1.$$

For the signed angular magnitude,

$$\alpha_n = |\operatorname{Arg} z_n|,$$

one obtains

$$\alpha_n = \frac{\pi - \alpha_{n-1}}{r_n}.$$

Proposition 14.2 (Angular synchronization). *Let (α_n) and $(\widetilde{\alpha}_n)$ solve the same variable-order recurrence with different initial values. Then*

$$|\alpha_n - \widetilde{\alpha}_n| = \frac{|\alpha_0 - \widetilde{\alpha}_0|}{N_n}.$$

In particular,

$$|\alpha_n - \widetilde{\alpha}_n| \le 2^{-n} |\alpha_0 - \widetilde{\alpha}_0| .$$

Proof. Subtract the two recurrences:

$$\alpha_n - \widetilde{\alpha}_n = -\frac{\alpha_{n-1} - \widetilde{\alpha}_{n-1}}{r_n}.$$

Iteration gives the result. □

Proposition 14.3 (Closed angular form for variable order). *Let*

$$N_n = \prod_{j=1}^{n} r_j.$$

Then

$$\alpha_n = \pi \sum_{j=1}^{n} \frac{(-1)^{n-j}}{r_j r_{j+1} \cdots r_n} + \frac{(-1)^n \alpha_0}{N_n}. \tag{50}$$

Equivalently,

$$\alpha_n = \frac{(-1)^n \alpha_0}{N_n} + \pi \left(\frac{1}{r_n} - \frac{1}{r_{n-1} r_n} + \cdots + \frac{(-1)^{n-1}}{N_n} \right).$$

Proof. Starting from $\alpha_n = (\pi - \alpha_{n-1})/r_n$, substitute recursively. Each substitution contributes one factor $-1/r_j$ to the preceding term. The final initial-condition term is therefore $(-1)^n \alpha_0 / N_n$, and the forcing terms give the displayed alternating sum. □

Remark 14.4 (Nonautonomous limiting angle) For constant order $r_n \equiv r$, the alternating sum is a geometric series and reduces to $\pi/(r+1)$ plus the transient term derived earlier. For a nonconstant order sequence there need not be a single autonomous fixed angle. What remains universal is exponential forgetting of the initial condition.

Remark 14.5 Synchronization does not by itself imply convergence to a single universal angle when (r_n) varies. It does imply that the nonautonomous dynamics asymptotically forgets its initial angular condition at a rate at least 2^{-n}.

14.1 Sharp inverse geometry for variable root orders

The product

$$N_n := \prod_{j=1}^{n} r_j$$

plays the exact role of r^n in the fixed-order inverse geometry.

Theorem 14.6 (Variable-order inverse levels). *Suppose that at stage j one takes all r_j roots in the inverse recurrence. Then the depth-n inverse level contains exactly N_n points, all with modulus*

$$|z_0|^{1/N_n},$$

and consecutive arguments separated by

$$\frac{2\pi}{N_n}.$$

Thus the level is a rotated regular N_n-gon.

Proof. The successive monomial degrees multiply, so the composed forward map has degree N_n. The depth-n inverse equation therefore has the form $w^{N_n} = c_n$ with $c_n \neq 0$. Its roots are exactly the N_n equally spaced points stated above. $\square$

Theorem 14.7 (Sharp variable-order Hausdorff asymptotic). *For the corresponding inverse sets,*

$$N_n d_H(A_n, S^1) \longrightarrow \sqrt{(\log|z_0|)^2 + \pi^2}. \tag{51}$$

In particular,

$$d_H(A_n, S^1) = O(N_n^{-1}).$$

Proof. The exact polygon argument from Section 9 applies after replacing r^n by N_n. The radius is

$$\exp\left(\frac{\log|z_0|}{N_n}\right)$$

and the angular half-gap is π/N_n. Taylor expansion yields the same quadratic leading constant as in the fixed-order case. Since $N_n \geq 2^n$, the expansion parameter tends to zero. $\square$

Corollary 14.8 (Variable-order Wasserstein convergence). *For the uniform variable-order inverse measure,*

$$W_1(\mu_n, m) = O(N_n^{-1}).$$

Proof. Use the nearest-vertex coupling. The radial error is $O(N_n^{-1})$ and the angular chord error is bounded by $2\sin(\pi/N_n) = O(N_n^{-1})$. $\square$

14.2 Periodic order sequences

If (r_n) is periodic of period q, then the angular recurrence is a periodically forced affine contraction. The q-step composition has slope

$$\frac{(-1)^q}{r_1 \cdots r_q},$$

whose modulus is less than one.

Proposition 14.9 (Unique periodic angular profile). *If $r_{n+q} = r_n$, there exists a unique q-periodic sequence (α_n^*) satisfying*

$$\alpha_n^* = \frac{\pi - \alpha_{n-1}^*}{r_n},$$

and every other solution synchronizes with it at the exact rate

$$|\alpha_n - \alpha_n^*| = \frac{C}{N_n}$$

for a constant C determined by the initial phase.

Proof. The q-step affine composition is a contraction and hence has a unique fixed point. Iterating the composition defines the periodic profile. The exact synchronization estimate follows from subtraction of the two nonautonomous recurrences, as in Section 14. $\square$

15 Logarithmic branches and branch defects

The root structure developed above is generated by logarithmic branches.

For $z \neq 0$,

$$\log z = \{\mathrm{Log}\, z + 2\pi i k : k \in \mathbb{Z}\}.$$

Hence

$$z^{1/r} = \left\{\exp\left(\frac{\mathrm{Log}\, z + 2\pi i k}{r}\right) : k \in \mathbb{Z}\right\}.$$

Only k modulo r changes the resulting root. Thus the infinite logarithmic branch index $k \in \mathbb{Z}$ is reduced to a finite branch index in

$$\mathbb{Z}/r\mathbb{Z}$$

at one root extraction.

This provides a basic bridge:

$$\boxed{\text{logarithmic sheet index} \longrightarrow \text{root branch index} \longrightarrow \text{finite branch history.}}$$

15.1 The logarithmic product defect

For $a, b \in \mathbb{C}^\times$, define

$$D(a,b) = \frac{\mathrm{Log}\, a + \mathrm{Log}\, b - \mathrm{Log}(ab)}{2\pi i}. \tag{52}$$

Proposition 15.1. *For all $a, b \in \mathbb{C}^\times$,*

$$D(a,b) \in \mathbb{Z}.$$

With the convention $\mathrm{Arg} \in (-\pi, \pi]$, *one has*

$$D(a,b) \in \{-1, 0, 1\}.$$

Proof. The exponential of the numerator in (52) is

$$\frac{ab}{ab} = 1.$$

Hence the numerator is an integral multiple of $2\pi i$.

Since each principal argument lies in $(-\pi, \pi]$, the difference between

$$\mathrm{Arg}\, a + \mathrm{Arg}\, b$$

and the principal representative of the argument of ab can only be -2π, 0, or 2π. □

Therefore

$$\mathrm{Log}(ab) = \mathrm{Log}\, a + \mathrm{Log}\, b - 2\pi i D(a,b). \tag{53}$$

This is a special instance of the unwinding-number philosophy used in the numerical analysis of complex functions; see [1, 3].

15.2 Complex powers

For principal powers,

$$z^\gamma = \exp(\gamma \operatorname{Log} z).$$

Equation (53) immediately gives

$$(ab)^\gamma = a^\gamma b^\gamma e^{-2\pi i\gamma D(a,b)}. \tag{54}$$

Thus the familiar real identity

$$(ab)^\gamma = a^\gamma b^\gamma$$

does not hold globally for arbitrary complex γ under principal powers.

It does hold whenever

$$\gamma D(a, b) \in \mathbb{Z}.$$

In particular, if $\gamma \in \mathbb{Z}$, the correction factor is one.

15.3 A corrected tangent-binomial identity

Where $\cos z \neq 0$,

$$\cos z(1 + i \tan z) = \cos z + i \sin z = e^{iz}.$$

For an integer n,

$$\cos^n z(1 + i \tan z)^n = e^{inz}$$

follows by ordinary algebra.

For arbitrary complex exponent γ, however, the principal-power version requires a branch correction. Define

$$D_z = \frac{\operatorname{Log}(\cos z) + \operatorname{Log}(1 + i \tan z) - iz}{2\pi i},$$

where all terms are defined.

Proposition 15.2 (Corrected principal-power identity). *For every complex γ for which the displayed principal powers are defined,*

$$\cos(z)^\gamma (1 + i \tan z)^\gamma = e^{i\gamma z} e^{2\pi i\gamma D_z}.$$

Proof. By definition,

$$\operatorname{Log}(\cos z) + \operatorname{Log}(1 + i \tan z) = iz + 2\pi i D_z.$$

Multiply by γ and exponentiate. □

This correction is important conceptually. It is not an obstacle to the underlying exponential identity; it is precisely the information lost when principal logarithmic representatives are substituted for a globally multivalued logarithm.

16 Complex logarithms and representations of π

The principal logarithm satisfies

$$\operatorname{Log}(i) = \frac{\pi i}{2}.$$

Consequently,

$$\pi = \frac{2 \operatorname{Log}(i)}{i}.$$

This identity is immediate from the definition of the principal logarithm. Its value in the present context is therefore conceptual rather than one of novelty: it illustrates that π appears as the fundamental angular scale associated with the principal logarithmic sheet.

More generally, whenever two logarithmic expressions differ by an integral multiple of $2\pi i$,

$$A - B = 2\pi i N,$$

one obtains

$$\pi = \frac{A - B}{2iN}, \qquad N \neq 0.$$

The mathematical interest of such formulas depends on their origin. A formula produced by simply rearranging the definition of Log should not be presented as a new constant-discovery mechanism. More substantial identities may arise when the integer N records a genuine branch defect or winding phenomenon.

17 Affine perturbations and the polynomial bridge

The monomial model also indicates exactly how an affine perturbation changes the full multivalued problem. Consider

$$z_{n+1} = \mathrm{Root}_r^{\mathrm{pr}}(az_n + b), \qquad a \neq 0. \tag{55}$$

The complete relation is

$$z_{n+1}^r = az_n + b,$$

so the predecessor satisfies

$$z_n = \frac{z_{n+1}^r - b}{a}. \tag{56}$$

Hence the full correspondence is the inverse relation of the degree-r polynomial

$$Q_{a,b}(z) = \frac{z^r - b}{a}.$$

Proposition 17.1 (Polynomial bridge for affine perturbations). *The full multivalued recurrence associated with (55) is inverse iteration of $Q_{a,b}$. The present negative-root model is the special case $a = -1$, $b = 0$.*

Proof. Equation (56) is precisely the predecessor relation for $Q_{a,b}$. □

The geometric consequences are immediate at the level of principle. When $b = 0$, the polynomial is conjugate to a monomial and the inverse levels are regular polygons. For $b \neq 0$, the Julia set is generally noncircular and the inverse tree no longer collapses to a single explicit regular polygon. General polynomial inverse iteration is then naturally described by the equilibrium measure and Julia geometry, while the exact monomial formula remains the solvable base case.

17.1 General deterministic branch selectors

For $z \neq 0$, every solution of $w^r = -z$ differs from the principal root by an rth root of unity. Thus every deterministic selector S with $S(z)^r = -z$ can be written

$$S(z) = \omega(z)F_r(z), \qquad \omega(z)^r = 1. \tag{57}$$

Here

$$\omega : \mathbb{C}^\times \to \mu_r, \qquad \mu_r = \{e^{2\pi ik/r} : 0 \leq k < r\}.$$

Proposition 17.2 (Representation of branch selectors). *A map $S : \mathbb{C}^\times \to \mathbb{C}^\times$ satisfies $S(z)^r = -z$ if and only if it has the representation (57) for some map $\omega : \mathbb{C}^\times \to \mu_r$.*

Proof. If $S(z)^r = F_r(z)^r = -z$, then $\omega(z) = S(z)/F_r(z)$ satisfies $\omega(z)^r = 1$. The converse is immediate. □

This gives a compact framework for measurable, piecewise, finite-state, Markov, or random branch selectors. The principal branch corresponds to the constant selector $\omega \equiv 1$.

18 Synthesis: one recurrence, several mathematically distinct limits

The preceding sections can now be assembled into a single logical picture. The point is not that every construction shares the same state space or the same notion of convergence. The point is that each construction can be reached from the same algebraic recurrence by changing exactly one modeling choice: branch selection, retention of all branches, symbolic coding, or probabilistic weighting.

18.1 Three levels of asymptotic geometry

For the fixed-order negative-root model there are three canonical asymptotic objects.

First, the principal function F_r selects one branch and produces a single deterministic orbit. For $z_0 \neq 0$ its radius satisfies

$$\log |z_n| = \frac{\log |z_0|}{r^n},$$

while its angle approaches the alternating pair

$$\pm\frac{\pi}{r+1}.$$

The asymptotic object is therefore a two-point periodic set.

Second, the full inverse correspondence retains all r^n branches at depth n. Its angular mesh is $2\pi/r^n$ and its radius differs from one by order r^{-n}. The asymptotic object is therefore the entire unit circle.

Third, a probability law assigns mass to the inverse branches. Uniform mass produces Haar measure, while a biased law produces a self-similar measure whose dimension is the source entropy divided by $\log r$. The asymptotic object is therefore not a set alone but a measure on the same geometric circle.

These three statements are compatible because they answer different questions:

one selected history	all histories	all histories + weights
$\Downarrow$	$\Downarrow$	$\Downarrow$
periodic orbit	circle	measure on the circle.

18.2 The phase parameter as a controlled deformation

The phase family makes the deterministic side substantially richer without destroying exact solvability. The radius remains governed by the same scalar contraction, so all nontrivial deformation occurs in the angle. After one root, the principal angle belongs to I_r, and the coordinate Ξ_r converts the branch-sensitive phase rule into

$$x_{n+1} = \left\{ \frac{x_n}{r} + \delta_{r,\beta} \right\}.$$

Thus the parameter β does not create a new arbitrary piecewise system: it moves the intercept of a fixed contraction ratio through a classical contracted-rotation family.

The resulting parameter space has a useful hierarchy. Open rational plateaus are structurally stable periodic regimes. Their endpoints are branch-boundary parameters at which symbolic labels may change. The exceptional irrational parameters form a zero-Hausdorff-dimension set, yet they support genuinely nonperiodic Cantor dynamics. The original negative-root parameter $\beta = \pi$ is especially transparent: it is the midpoint of a period-two window rather than an isolated exceptional value.

18.3 Why the exact reduction is mathematically useful

The contracted-rotation reduction is more than a change of notation. It separates three sources of complexity that would otherwise be mixed together:

$$\boxed{\text{radial contraction} \quad + \quad \text{branch reduction} \quad + \quad \text{symbolic asymptotics.}}$$

The first is completely explicit and never causes bifurcation. The second is encoded by the fractional-part operation. The third is organized globally by rotation number and, in the irrational regime, by the associated continued fraction and symbolic structure. This decomposition makes it possible to transfer established one-dimensional theorems without pretending that the complex-root problem itself is a new abstract class of map.

18.4 Convergence must always be labeled

The paper contains several precise modes of convergence, and keeping their names visible prevents a recurrent logical mistake. For the deterministic principal orbit one has convergence of even and odd subsequences to a two-cycle, not generally convergence of the full sequence. For inverse levels one has Hausdorff convergence of finite sets. For uniform inverse measures one has weak convergence and an explicit Wasserstein bound. For the symbolic Bernoulli series one has almost-sure convergence. For the chronological random recursion one generally has convergence in distribution. For variable order one has synchronization at a rate controlled by the cumulative degree.

Accordingly, statements of the form "the infinite radical converges" should always be replaced by a statement specifying the object and the topology or mode of convergence. In the present model this is not pedantry: the distinction is exactly what permits periodic, circular, fractal, and probabilistic limits to coexist without contradiction.

19 Literature context and status of the contributions

The mathematical structures appearing here belong to several existing literatures.

19.1 Complex iterated radicals

Complex iterated radicals were studied well before the present work. Walker [14] discusses the convergence problem and points to earlier work of Ogilvy [11], Rohde [12], and Gerber [13].

The present paper therefore does not claim priority for the general subject of complex recursive radicals.

What is distinctive in the present formulation is the systematic separation of principal deterministic dynamics from the complete inverse correspondence and the explicit development of both for

$$P_r(z) = -z^r.$$

19.2 Contracted rotations

The phase family is exactly conjugate, after one root extraction and the natural post-root coordinate change, to the circle contraction

$$x \mapsto \{\lambda x + \delta\}, \qquad \lambda = \frac{1}{r}.$$

Laurent and Nogueira develop the rotation-number and arithmetic theory of this family, including its description by Hecke–Mahler series and the consequence that algebraic parameters force rational rotation number [5]. The more recent work of Gaivão, Laurent, and Nogueira treats two-interval piecewise-affine maps and obtains further exact rotation-number and arithmetic relations [6]. Gaivão proved in 2025 that the exceptional parameter set for general interval piecewise-affine contractions has Hausdorff dimension zero [7].

The relevance here is exact rather than analogical: the coordinate construction in Section 6 embeds the phase-perturbed radical family into this established theory. The finite branch-admissibility formulas developed in this paper complement the global rotation-number classification.

19.3 Inverse iteration and monodromy

The inverse tree of a polynomial is a classical object in complex dynamics. Iterated monodromy groups provide a general framework for its algebraic and topological structure; see Nekrashevych [10]. The power map provides a particularly explicit case. Since $-z^r$ is linearly conjugate to z^r, the present negative-root tree inherits that classical structure.

The contribution here is therefore an explicit identification of recursive branch histories with the rooted-tree structure, together with a declared base-r terminal coding once a branch-labeling convention has been fixed.

19.4 Self-similar measures and dimension

The entropy–dimension relation developed in Section 11 belongs to standard symbolic and self-similar measure theory. The foundational existence theory for invariant compact sets and measures under contractions is due to Hutchinson [4], and modern fractal dimension theory is treated extensively by Falconer [2].

The role of the present example is that the branch probabilities are not artificially introduced: they arise directly from assigning probabilities to the finitely many possible complex roots at each recursive step.

19.5 Logarithmic branch defects

The failure of unrestricted principal-logarithm product identities is classical. The unwinding-number framework of Corless and Jeffrey and its subsequent exposition in numerical complex analysis provide systematic language for these defects [1, 3].

The present paper uses this mechanism to correct complex-power factorizations and to connect logarithmic branch information with the finite branch indices used in recursive roots.

20 A classification of statements in this paper

For clarity, the main mathematical statements can be classified as follows.

Result	**Status**	**Role in this paper**
Principal logarithm and root theory	Classical	Background and conventions
Principal negative-root angular contraction	Direct explicit analysis	Central exact model
Unique nonzero attracting two-cycle	Explicit theorem for this model	Core deterministic result
Phase normalization and relation to contracted rotations	Reformulation/synthesis	Connects root dynamics to existing theory
Exact contracted-rotation conjugacy	Direct explicit derivation	Identifies the phase family exactly with the standard $1/r$ contraction
Rotation-number and arithmetic consequences	Imported theorems applied through conjugacy	Periodicity, irrational regimes, exceptional parameters, and arithmetic
Julia-set and equilibrium interpretation	Classical theory specialized explicitly	Embeds inverse polygons in polynomial dynamics and potential theory
Transfer-operator formulation	Direct explicit derivation	Explains exact Fourier cancellation operator-theoretically
Multifractal frequency spectrum	Direct symbolic derivation	Refines entropy–dimension to frequency level sets
Chronological random branch dynamics	Direct formulation	Separates chronological paths, reversed symbolic series, and distributional limits
Variable-order inverse geometry	Direct extension	Extends sharp polygon and Wasserstein rates to cumulative degree N_n
General branch-selector representation	Direct algebraic representation	Unifies deterministic branch choices
Symbolic periodic-word formula	Direct explicit derivation	Finite exact periodicity algorithm
Inverse polygons of $-z^r$	Direct consequence of monomial inversion	Core multivalued geometry
Sharp Hausdorff asymptotic	Explicit theorem for the inverse levels	Quantitative geometric result
Base-r branch coding	Classical coding used explicitly	Bridge between branches and inverse trees
Monodromy/odometer framework	Classical theory	Interpretative structural connection
Entropy–dimension formula	Classical symbolic measure theory applied here	Probabilistic branch geometry
Variable-order synchronization	Direct extension	Nonautonomous direction
Logarithmic branch-defect correction	Classical mechanism applied explicitly	Corrects principal-power identities

This classification is intended to prevent two opposite mistakes: claiming novelty for classical mathematics, and overlooking the value of a mathematically useful synthesis.

21 Open problems

The exact solvability of the monomial model suggests several natural extensions.

The exact contracted-rotation conjugacy removes the need to ask whether a canonical rotation-type factor exists: the factor is the explicit coordinate Ξ_r. The remaining questions concern the arithmetic and combinatorial structure visible after that factor has been constructed.

Problem 21.1 (Explicit mode-locking tongues and endpoints) For fixed r, determine closed-form formulas for all rational rotation-number plateaus as intervals in the original phase parameter β, including their endpoint conventions under $\operatorname{Arg} \in (-\pi, \pi]$. The period-one and period-two windows proved above suggest an inductive finite-word approach.

Problem 21.2 (Symbolic classification of the exceptional phase set) For irrational $\rho_r(\beta)$, describe the admissible symbolic language in root coordinates directly, rather than only through the abstract contracted rotation. In particular, relate continued-fraction data for $\rho_r(\beta)$ to the asymptotic frequencies of principal-branch reductions.

Problem 21.3 (General deterministic branch-selection rules) Classify deterministic maps

$$S : \mathbb{C}^\times \longrightarrow \mathbb{C}^\times$$

such that

$$S(z)^r = -z.$$

Which measurable, continuous-away-from-cuts, or symbolic selection rules produce fixed points, periodic orbits, or more complicated dynamics?

Problem 21.4 (Affine perturbations) Study

$$z_{n+1} = \operatorname{Root}_r^{\mathrm{pr}}(az_n + b).$$

Which exact structures of the monomial model survive when $b \neq 0$?

Problem 21.5 (Dependent branch processes) Replace the iid branch measure by a stationary Markov process or a more general shift-invariant measure on Σ_r. Determine how entropy and dimension interact with the angular coding.

Problem 21.6 (Monodromy versus recursive branch history) Formulate a precise general theorem relating analytic continuation around branch points to transformations of recursive branch histories for multivalued analytic correspondences beyond the monomial case.

22 Conclusion

The recurrence

$$z_{n+1}^r = -z_n$$

contains several mathematically different systems. Treating them as one object obscures their structure.

For the principal branch, the dynamics is deterministic and exactly solvable. The modulus converges to one, the signed angular magnitude is governed by an affine contraction, and every nonzero orbit approaches the unique two-cycle

$$\left\{e^{i\pi/(r+1)}, e^{-i\pi/(r+1)}\right\}.$$

For the full multivalued operation, the same recurrence becomes inverse dynamics of

$$P_r(z) = -z^r.$$

Its depth-n preimages form a regular r^n-gon. The polygonal levels converge in Hausdorff distance to the unit circle, with an exact asymptotic constant.

Branch histories provide the missing combinatorial layer. They are finite base-r words at finite depth and infinite symbolic sequences at infinite depth. Through the linear conjugacy between $-z^r$ and z^r, these histories also connect naturally to the classical rooted trees and monodromy structures associated with power maps.

The phase-perturbed family supplies a unifying dynamical reduction beyond the original two-cycle. After one root extraction, the angular coordinate is exactly conjugate to a contracted rotation, so its global complex ω-limit set is the unit-circle image of the corresponding contracted-rotation attractor. The original $\beta = \pi$ case is the center of an explicit period-two mode-locking window, while other rational phases yield periodic regimes and the exceptional irrational phases lead to Cantor-type asymptotics.

Probability adds a further layer. Uniform branch selection gives Haar measure, while biased branch selection gives Bernoulli measures whose dimension is controlled exactly by the entropy of the branch source. The chronological random recursion must, however, be distinguished from the almost-surely convergent reversed symbolic series: in general its natural limit statement is convergence in distribution to the same Bernoulli law.

Finally, logarithmic branches provide the analytic mechanism behind root branches and explain why naive principal-power identities require integer correction terms.

The principal conclusion is therefore not that recursive complex roots create a new kind of number system. Rather, they provide an unusually explicit laboratory in which several classical structures meet:

$$\boxed{\begin{array}{c}\text{multivalued analysis} + \text{branch selection} + \text{symbolic dynamics} \\ + \text{inverse iteration} + \text{monodromy} + \text{geometric convergence} \\ + \text{probability and entropy.}\end{array}}$$

The negative-root monomial family is simple enough to solve exactly, yet rich enough to make the distinction between these structures mathematically visible.

A A computational verification protocol

The principal results of this paper are proved analytically. Numerical computation can nevertheless provide useful independent verification.

A.1 Principal iteration

Given precision P decimal digits and $z_0 \in \mathbb{C}$, compute

$$z_{n+1} = \exp\left(\frac{1}{r}\left[\log|-z_n| + i\,\mathrm{Arg}(-z_n)\right]\right).$$

At each step record

$$\epsilon_n^{(\mathrm{rad})} = \left|\log|z_n| - \frac{\log|z_0|}{r^n}\right|$$

and

$$\epsilon_n^{(\mathrm{ang})} = \left|\alpha_n - \frac{\pi}{r+1} - \left(-\frac{1}{r}\right)^n \left(\alpha_0 - \frac{\pi}{r+1}\right)\right|.$$

Both should be numerically consistent with the selected arithmetic precision.

A.2 Inverse polygons

At depth n, solve

$$w^{r^n} = (-1)^{1+r+\cdots+r^{n-1}} z_0.$$

The expected diagnostics are:

1. exactly r^n distinct roots;
2. equal moduli within numerical tolerance;
3. angular gaps approximately $2\pi/r^n$;
4. Hausdorff distance consistent with (34).

A.3 Symbolic periodicity

For a word

$$(k_0, \ldots, k_{p-1}),$$

compute (17). Then iterate the affine recurrence for p steps and test all inequalities (18). Only after admissibility is verified should the candidate be classified as a genuine principal periodic orbit.

B Proof-dependency map and integrity checklist

This appendix records the logical dependencies among the principal statements. It is included to make the manuscript auditable: a reader can verify the main claims without having to infer which earlier facts are being used implicitly.

B.1 Dependency map

Statement	**Dependencies**
Radial law	Definition of principal root; real part of $\operatorname{Log} z$.
Principal angular map	Principal-argument convention; explicit reduction of $-z$.
Scalar angular recurrence	Principal angular map; absolute value of the signed angle.
Exact angular solution	Affine recurrence $\alpha \mapsto (\pi - \alpha)/r$.
Two-cycle theorem	Radial convergence + angular convergence + sign alternation.
Periodic classification	Periodicity of radius + contraction of angular magnitude.
Inverse-level formula	Closed form for P_r^n.
Regular-polygon theorem	Fundamental theorem on roots of a nonzero complex number.
Hausdorff formula	Exact polygon radius + maximal half-gap π/r^n.
Sharp Hausdorff rate	Taylor expansions of e^x and $\cos x$.
Uniform measure limit	Root-of-unity cancellation + radial convergence.
Exact contracted-rotation conjugacy	Post-root invariant interval + affine coordinate change.
Rotation-number arithmetic	Exact conjugacy + arithmetic theorems.
Julia-set theorem	Radial escape law + linear conjugacy with z^r.
Equilibrium interpretation	Explicit inverse polygons + classical preimage theory.
Transfer-operator identities	Root-of-unity sums on Fourier characters.
Entropy–dimension formula	Bernoulli cylinder probabilities + strong law + r-adic coding.
Multifractal spectrum	Frequency counting + cylinder diameters.
Random branch limit	Contractive affine recursion + reversal in law; no general pathwise limit.
Variable-order synchronization	Subtraction of two affine recurrences.
Variable-order Hausdorff rate	Regular inverse polygon geometry + Taylor expansion.
Logarithmic defect formula	Definition of principal logarithm + $e^{2\pi ik} = 1$.

B.2 Convention checklist

Before using any formula involving a principal argument, verify the following:

(1) the representative lies in $(-\pi, \pi]$;

(2) the input is nonzero whenever Log or Arg is invoked;

(3) any multiplication by a phase has been reduced before a principal root is extracted;

(4) any branch integer appearing in a symbolic itinerary satisfies its finite admissibility inequalities;

(5) endpoint cases are checked with the same half-open convention used in the definition.

B.3 Independence of numerical verification

The computations described in Appendix A are deliberately diagnostic rather than deductive. They test exact identities at finite precision but do not serve as substitutes for the analytic proofs. In particular, a numerical trajectory can conceal a branch-cut error if the precision is insufficient or if the principal-argument convention of the software differs from the convention of this paper. The analytic statements therefore remain primary.

Principal result

Minimal verification protocol. A trustworthy implementation should independently check: (i) the radial law, (ii) the scalar angular law, (iii) the inverse equation at several depths, (iv) angular equal-spacing, (v) the exact Hausdorff expression, (vi) the finite admissibility inequalities for any claimed symbolic periodic orbit, (vii) the exact contracted-rotation coordinate change, and (viii) representative rational and irrational phase regimes. Passing these checks is strong evidence of faithful implementation, but the proofs and cited external theorems in the body of the paper are what establish the mathematical results.

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